

Mendelson Notes

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1 The Propositional Calculus

1.1 Propositional Connectives: Truth Tables

Truth-functional Connectives (assume capital letters represent sentences)

Negation:

A	$\neg A$
T	F
F	T

Conjunction / And:

A	B	$A \wedge B$
T	T	T
F	T	F
T	F	F
F	F	F

Disjunction / Inclusive Or:

A	B	$A \vee B$
T	T	T
F	T	T
T	F	T
F	F	F

Conditional / If-then:

A	B	$A \Rightarrow B$
T	T	T
F	T	T
T	F	F
F	F	T

Biconditional / If-and-only-if:

<i>A</i>	<i>B</i>	$A \Leftrightarrow B$
<i>T</i>	<i>T</i>	<i>T</i>
<i>F</i>	<i>T</i>	<i>F</i>
<i>T</i>	<i>F</i>	<i>F</i>
<i>F</i>	<i>F</i>	<i>T</i>

For Mendlelson's notation:

1. Capital italic letters (sometimes w/ subscripts) are statement forms
2. if $\mathcal{B}$ and $\mathcal{C}$ are statement forms, we can modify them with propositional connectives to form other statements

I don't feel the need to recreate the large truth tables, as they are clear enough in the text.

Note: If there are n distinct letters in a statement form, there are 2^n possible assignments of truth values to the statement letters, meaning 2^n rows in the truth table.

1.2 Tautologies

2 First-Order Logic and Model Theory

2.1 Quantifiers

If $P(x)$ asserts that x has the property P , then $(\forall x)P(x)$ means that property P holds for all x , or, in other words, that everything has the property P .

On the other hand, $(\exists x)P(x)$ means that some x has the property P —that is, that there is at least one object having the property P .

Symbols for FOL:

- Commas, parentheses, $\neg$, $\Rightarrow$, $\forall$, $\exists$, and:
- Individual variables: $x_1, x_2, \dots, x_n, \dots$
- Individual constants: $a_1, a_2, \dots, a_n, \dots$
- Predicate letters: A_k^n
- Function letters: f_k^n

n is the number of arguments, k is the indexing number to distinguish various functions and predicates with the same number of arguments. Functions applied to variables and individual constants generate terms.

Variables vs. Terms:

1. Variables and individual constants are terms.
2. If f_k^n is a function letter and $t_1, \dots, t_n$ are terms, then $f_k^n(t_1, \dots, t_n)$ is a term.
3. An expression is a term iff it can be shown to be a term on the basis of conditions 1 and 2.

Well-formed formulas:

$A_k^n(t_1, \dots, t_n)$ is an atomic formula. All wfs can be defined as follows:

1. Every atomic formula.
2. If $\mathcal{B}$ and $\mathcal{C}$ are wfs and y is a variable, $(\neg\mathcal{B})$, $(\mathcal{B} \Rightarrow \mathcal{C})$, and $((\forall y)\mathcal{B})$ are wfs.
3. An expression is a wf iff it can be shown to be a wf on the basis of conditions 1 and 2.

Note that if $\mathcal{B}$ does not contain y , then $((\forall y)\mathcal{B})$ means the same thing as $\mathcal{B}$. Also, note that we can define $((\exists y)\mathcal{B})$ in terms of $\neg((\forall y)(\neg\mathcal{B}))$

Parenthesis:

An example of restoring parenthesis:

1. $(\forall x_1)A_1^1(x_1) \Rightarrow A_1^2(x_2, x_1)$
2. $((\forall x_1)A_1^1(x_1)) \Rightarrow A_1^2(x_2, x_1)$
3. $((\forall x_1)A_1^1(x_1)) \Rightarrow A_1^2(x_2, x_1)$

Bound and Free Variables:

An occurrence of a variable x is *bound* for a wf $\mathcal{B}$ if it is either the x in $(\forall x)$ or it is within the scope of a $(\forall x)$ in $\mathcal{B}$.

For example:

1. $A_1^2(x_1, x_2)$
2. $A_1^2(x_1, x_2) \Rightarrow (\forall x_1)A_1^1(x_1)$

In example 1, the single occurrence of x_1 is free. In example 2, the first occurrence of x_1 is free, but the other two are bound. Check the textbook for more examples.

Sometimes we indicate that some variables $x_{i_1}, \dots, x_{i_k}$ are free variables in a wf $\mathcal{B}$ by writing $\mathcal{B}(x_{i_1}, \dots, x_{i_k})$. This does not mean that $\mathcal{B}$ contains these variables as free variables, nor does it mean that $\mathcal{B}$ does not contain other free variables. This notation is convenient because we can then agree to write as $\mathcal{B}(t_1, \dots, t_k)$ the result of substituting in $\mathcal{B}$ the terms $t_1, \dots, t_k$ for all free occurrences (if any) of $x_{i_1}, \dots, x_{i_k}$, respectively.

If $\mathcal{B}$ is a wf and t is a term, then t is said to be *free* for x_i in $\mathcal{B}$ if no free occurrence of x_i in $\mathcal{B}$ lies within the scope of any quantifier $(\forall x_j)$, where x_j is a variable in t . This concept of t being free for x_i for a wf $\mathcal{B}(x_i)$ will be important later. This means that, if t is substituted for all free occurrences (if any) of x_i in $\mathcal{B}(x_i)$, no occurrence of a variable in t becomes a bound occurrence in $\mathcal{B}(t)$.

Some important facts:

1. A term that contains no variables is free for any variable in any wf.
2. A term t is free for any variable in $\mathcal{B}$ if none of the variables of t is bound in $\mathcal{B}$.
3. x_i is free for x_i in any wf.
4. Any term is free for x_i in $\mathcal{B}$ if $\mathcal{B}$ contains no free occurrences of x_i .

Exercise 2.6: Is the term $f_1^2(x_1, x_2)$ free for x_1 in the following wfs?

- | | |
|---|--------|
| a. $A_1^2(x_1, x_2) \Rightarrow (\forall x_2)A_1^1(x_2)$ | A: yes |
| b. $((\forall x_2)A_1^2(x_2, a_1)) \vee (\exists x_2)A_1^2(x_1, x_2)$ | A: no |
| c. $(\forall x_1)A_1^2(x_1, x_2)$ | A: yes |
| d. $(\forall x_2)A_1^2(x_1, x_2)$ | A: no |
| e. $(\forall x_2)A_1^1(x_2) \Rightarrow A_1^2(x_1, x_2)$ | A: yes |

Translating English sentences to wfs:

One example: *Some people respect everyone:* $(\exists x)(P(x) \wedge (\forall y)(P(y) \Rightarrow R(x, y)))$ where $P(x)$ means x is a person and $R(x, y)$ means x respects y . More examples for this section on page 51 of the book, page 7 of chapter 2 pdf.

2.2 FOL's and Their Interpretations: Satisfiability and Truth: Models

We use the following symbols for our first-order Language $\mathcal{L}$.

- a. The propositional connectives $\neg$ and $\Rightarrow$, as well as $\forall$.
- b. Punctuation marks: "(", ")", and ", ,".
- c. Denumerably many individual variables $x_1, x_2, \dots$

- d. A finite or denumerable, possibly empty, set of function letters.
- e. A finite or denumerable, possibly empty, set of individual constants.
- f. A nonempty set of predicate letters.

A *term* of $\mathcal{L}$ means a term whose symbols are symbols of $\mathcal{L}$.

A *wf* of $\mathcal{L}$ means a wf whose symbols are symbols of $\mathcal{L}$.

«**Note**» denumerable = finite or countably infinite (e.g. can be mapped one-to-one with $\mathbb{N}$).

Now we define an interpretation M of $\mathcal{L}$:

- a. A nonempty set D , called the *domain* of the interpretation.
- b. For each predicate letter A_j^n of $\mathcal{L}$, an assignment of an n -place relation $(A_j^n)^M$ in D .
- c. For each function letter f_j^n of $\mathcal{L}$, an assignment of an n -place operation $(f_j^n)^M$ in D (that is, a function from D^n to D).
- d. For each individual constant a_i of $\mathcal{L}$, an assignment of some fixed element $(a_i)^M$ of D .

An n -place relation in D can be thought of as a subset of D^n , the set of all n -tuples of elements in D . For a given interpretation of $\mathcal{L}$, a wf without free variables, e.g. a closed wf, is either true or false. A wf with some free variables may be satisfied, i.e. true, for some values of the domain and not for others.

Satisfaction

A denumerable sequence $s = (s_1, s_2, s_3, \dots)$ satisfies a wf $\mathcal{B}$ that has $x_{j_1}, x_{j_2}, \dots, x_{j_n}$ as free variables (where $j_1 < j_2 < \dots < j_n$) if the n -tuple $\langle s_{j_1}, s_{j_2}, \dots, s_{j_n} \rangle$ satisfies $\mathcal{B}$ in the usual sense. «**clarify this**»

For example, a denumerable sequence $(s_1, s_2, s_3, \dots)$ of objects in the domain of an interpretation M will turn out to satisfy the wf $A_1^2(x_2, x_5)$ iff the ordered pair, $\langle s_2, s_5 \rangle$ is in the relation $(A_1^2)^M$ assigned to the predicate letter A_1^2 by the interpretation M .

Let M be an interpretation of a language $\mathcal{L}$ and let D be the domain of M . Let Σ be the set of all denumerable sequences of elements of D . For a wf $\mathcal{B}$ of $\mathcal{L}$, we shall define what it means for a sequence $s = (s_1, s_2, \dots)$ in Σ to satisfy $\mathcal{B}$ in M . As a preliminary step, for a given $s \in \Sigma$ we shall define a function s^* that assigns to each term t of $\mathcal{L}$ an element $s^*(t)$ in D .

- 1. If t is a variable x_j , let $s^*(t)$ be s_j .
- 2. If t is an individual constant a_j , then $s^*(t)$ is the interpretation $(a_j)^M$.
- 3. If f_k^n is a function letter, $(f_k^n)^M$ is the corresponding operation in D , and $t_1, \dots, t_n$ are terms, then

$$s^*(f_k^n(t_1, \dots, t_n)) = (f_k^n)^M(s^*(t_1), \dots, s^*(t_n))$$

Also, $s^*(t)$ is the element of D obtained by substituting, for each j , a name of s_j for all occurrences of x_j in t and then performing the operations of the interpretation corresponding to the function letters of t .

Now for an inductive definition of satisfaction.

1. If $\mathcal{B}$ is an atomic wf $A_k^n(t_1, \dots, t_n)$ and $(A_k^n)^M$ is the corresponding n -place relation of the interpretation, then a sequence $s = (s_1, s_2, \dots)$ satisfies $\mathcal{B}$ iff $(A_k^n)^M(s^*(t_1), \dots, s^*(t_n))$, e.g. if the n -tuple $\langle s^*(t_1), \dots, s^*(t_n) \rangle$ is in the relation $(A_k^n)^M$.
2. s satisfies $\mathcal{B}$ iff s does not satisfy $\neg\mathcal{B}$.
3. s satisfies $\mathcal{B} \Rightarrow \mathcal{C}$ iff s does not satisfy $\mathcal{B}$ or s satisfies $\mathcal{C}$.
4. s satisfies $(\forall x_i)\mathcal{B}$ iff every sequence that differs from s in at most the i th component satisfies $\mathcal{B}$.

What this also means is that a sequence $s = (s_1, s_2, \dots)$ satisfies a wf $\mathcal{B}$ iff, when, for each i , we replace all free occurrences of x_i (if any) in $\mathcal{B}$ by a symbol representing s_i , the resulting proposition is true under the given interpretation.

Truth and Falsity of wfs in Interpretations

1. A wf $\mathcal{B}$ is true for the interpretation M ($\models_M \mathcal{B}$) iff every sequence in Σ satisfies $\mathcal{B}$.
2. $\mathcal{B}$ is said to be false for M iff no sequence in Σ satisfies $\mathcal{B}$.
3. An interpretation M is said to be a *model* for a set Γ of wfs iff every wf in Γ is true for M .

This leads to the following properties of truth, falsity, and satisfaction. I-V are very obvious so I'm skipping those (e.g. they are specifying how logical operators from chapter 1 behave as expected in this new context).

- VI. $\models_M$ iff $\models_M (\forall x_i)\mathcal{B}$. This result can be extended. By the closure of $\mathcal{B}$ we mean the closed wf obtained from $\mathcal{B}$ by prefixing in universal quantifiers those variables, in order of descending subscripts, that are free in $\mathcal{B}$. If $\mathcal{B}$ has no free variables, its closure is $\mathcal{B}$. For example, if $\mathcal{B}$ is $A_1^2(x_2, x_5) \Rightarrow \neg(\exists x_2)A_1^3(x_1, x_2, x_3)$ then its closure is $(\forall x_5)(\forall x_3)(\forall x_2)(\forall x_1)\mathcal{B}$. A wf $\mathcal{B}$ is true iff its closure is true.
- VII. Every instance of a tautology is true for any interpretation. To prove this, show that all instances of the axioms of the system L are true and then use (III) and Prop 1.14.
- VIII. If the free variables of a wf $\mathcal{B}$ occur in the list $x_{i_1}, \dots, x_{i_k}$ and if the sequences s and s' have the same components in the i_1 th, $\dots$, i_k th places, then s satisfies $\mathcal{B}$ iff s' satisfies $\mathcal{B}$.
- IX. If $\mathcal{B}$ is a closed wf of $\mathcal{L}$, then either $\models_M \mathcal{B}$ or $\models_M \neg\mathcal{B}$ for some interpretation M , e.g. either $\mathcal{B}$ is true or $\mathcal{B}$ is false for M . If $\mathcal{B}$ is not closed (e.g. contains free variables) then $\mathcal{B}$ may be true, false, or neither for some interpretation M . For example, if $\mathcal{B}$ is $A_1^2(x_1, x_2)$ and we consider an interpretation in which the domain is the set of integers and $A_1^2(y, z)$ is interpreted as $y < z$, then $\mathcal{B}$ is only satisfied by sequences $s = (s_1, s_2, \dots)$ such that $s_1 < s_2$. This means that $\mathcal{B}$ is not satisfied by all sequences, hence it is neither true nor false for all interpretations.
- X. Assume that t is free for x_i in $\mathcal{B}(x_i)$. Then $(\forall x_i)\mathcal{B}(x_i) \Rightarrow \mathcal{B}(t)$ is true for all interpretation. The proof of (X) is based upon lemmas 1 and 2 from the text (page 59 & 60).
- XI. If $\mathcal{B}$ does not contain x_i free, then $(\forall x_i)(\mathcal{B} \Rightarrow \mathcal{C}) \Rightarrow (\mathcal{B} \Rightarrow (\forall x_i)\mathcal{C})$ is true for all interpretations.

Definitions

Logically valid: A wf $\mathcal{B}$ which is true for every interpretation.

Satisfiable: A wf $\mathcal{B}$ for which there is an interpretation which satisfies $\mathcal{B}$ for at least one sequence. Note that if $\mathcal{B}$ is closed, it is either satisfied by all sequences or by none. So, $\mathcal{B}$ is satisfiable iff $\mathcal{B}$ is true for some interpretation.

Contradictory: A wf $\mathcal{B}$ which is false for every interpretation, e.g. $\neg\mathcal{B}$ is logically valid.

Logically imply: $\mathcal{B}$ logically implies $\mathcal{C}$ iff, in every interpretation, every sequence which satisfies $\mathcal{B}$ also satisfies $\mathcal{C}$. Moreover, $\mathcal{C}$ is a *logical consequence* of a set Γ of wfs iff, in every interpretation, every sequence that satisfies every wf in Γ also satisfies $\mathcal{C}$.

Logically equivalent: $\mathcal{B}$ and $\mathcal{C}$ logically imply each other.

2.3 First-Order Theories

Note that the truth table method for checking whether a given statement is a tautology fails for Chapter 2, as now this would require us to check the truth of a given wf for interpretations with arbitrarily large or infinite domains.

Let $\mathcal{L}$ be a FOL. A *first-order theory* in the language $\mathcal{L}$ will be a formal theory K whose symbols and wfs are the symbols and wfs of $\mathcal{L}$ and whose axioms and rules of inference are specified as follows by 2.3.1-2.3.3.

Axioms of K are either logical or proper (nonlogical). For most of this chapter we consider K with no proper axioms, e.g. a Predicate Calculus.

2.3.1 Logical Axioms

If $\mathcal{B}$, $\mathcal{C}$, and $\mathcal{D}$ are wfs of $\mathcal{L}$, then the following are logical axioms of K :

- (A1) $\mathcal{B} \Rightarrow (\mathcal{C} \Rightarrow \mathcal{B})$
- (A2) $(\mathcal{B} \Rightarrow (\mathcal{C} \Rightarrow \mathcal{D})) \Rightarrow ((\mathcal{B} \Rightarrow \mathcal{C}) \Rightarrow (\mathcal{B} \Rightarrow \mathcal{D}))$
- (A3) $(\neg\mathcal{C} \Rightarrow \neg\mathcal{B}) \Rightarrow ((\neg\mathcal{C} \Rightarrow \mathcal{B}) \Rightarrow \mathcal{C})$
- (A4) $(\forall x_i)\mathcal{B}(x_i) \Rightarrow \mathcal{B}(t)$ if $\mathcal{B}(x_i)$ is a wf of $\mathcal{L}$ and t is a term of $\mathcal{L}$ that is free for x_i in $\mathcal{B}(x_i)$.
Note that t may be identical with x_i so that all wfs $(\forall x_i)\mathcal{B} \Rightarrow \mathcal{B}$ are axioms by virtue of axiom (A4).
- (A5) $(\forall x_i)(\mathcal{B} \Rightarrow \mathcal{C}) \Rightarrow (\mathcal{B} \Rightarrow (\forall x_i)\mathcal{C})$ if $\mathcal{B}$ contains no free occurrences of x_i .

Explaining the restrictions on (A4) and (A5):

(A4): Consider this example where t is not free for x_i in $\mathcal{B}(x_i)$. Let $\mathcal{B}(x_1)$ be $\neg(\forall x_2)A_1^2(x_1, x_2)$ and let t be x_2 . Now, consider the following (illegal) instance of (A4):

$$(\Delta) \quad (\forall x_1)(\neg(\forall x_2)A_1^2(x_1, x_2)) \Rightarrow \neg(\forall x_2)A_1^2(x_2, x_2)$$

Now take an interpretation as any domain with at least two members and let A_1^2 stand for the identity relation. Then, the antecedent of (Δ) is true and the consequent is false.

(A5): Consider this example where x_i is free in $\mathcal{B}$. Let $\mathcal{B}$ and $\mathcal{C}$ both be $A_1^1(x_1)$. Now, consider the following (illegal) instance of axiom (A5):

$$(\Delta\Delta) \quad (\forall x_1)(A_1^1(x_1) \Rightarrow A_1^1(x_1)) \Rightarrow (A_1^1(x_1) \Rightarrow (\forall x_1)A_1^1(x_1))$$

The antecedent of $(\Delta\Delta)$ is logically valid. Now take as domain the set of integers and let $A_1^1(x)$ mean that x is even. Then the consequent is false. So, any sequence $s = (s_1, s_2, \dots)$ for which s_1 is even does not satisfy the consequent of $(\Delta\Delta)$, making it false for this interpretation.

2.3.2 Proper Axioms

These vary for different theories. As stated before, a theory with no proper axioms is a *predicate calculus*.

2.3.3 Rules of Inference

1. Modus ponens (MP): $\mathcal{C}$ follows from $\mathcal{B}$ and $\mathcal{B} \Rightarrow \mathcal{C}$.
2. Generalization (Gen): $(\forall x_i)\mathcal{B}$ follows from $\mathcal{B}$.

Definition

If K is a FOT of $\mathcal{L}$, a *model* of K is an interpretation of $\mathcal{L}$ for which all the axioms of K are true. Every theorem of K is true in every model of K .

2.4 Properties of First-Order Theories

«Note» For this chapter assume that $\vdash$ means $\vdash_K$.

Proposition 2.1

Every wf $\mathcal{B}$ of K that is an instance of a tautology is a theorem of K , and it may be proved using only axioms (A1)-(A3) and MP.

Proof

$\mathcal{B}$ arises from a tautology $\mathcal{J}$ by substitution. Because $\mathcal{J}$ is a tautology, by Proposition 1.14 we know there is a proof of $\mathcal{J}$ in L . In this proof, make the same substitution of wfs of K for statement letters as were used in obtaining $\mathcal{B}$ from $\mathcal{J}$, and, for all statement letters in the proof that do not occur in $\mathcal{J}$, substitute an arbitrary wf of K . The resulting sequence of wfs is a proof of $\mathcal{B}$, which only uses (A1)-(A3) and MP because these were the only axioms and inference rules available in Chapter 1 for our propositional calculus. Usage of this will be indicated by writing "Tautology."

Proposition 2.2

Every theorem of a first-order predicate calculus is logically valid.

Proof

We know that (A1)-(A5) are logically valid and that our inference rules, MP and Gen, preserve logical validity. So, any wf obtained from a sequence of these wfs and rules of inference must also be logically valid (page 69 for more specific details).

Definition

A theory K is *consistent* if no wf $\mathcal{B}$ and $\mathcal{B}$ are both provable in K . A theory is *inconsistent* otherwise.

Corollary 2.3

Any first-order predicate calculus is consistent

Proof

First, consider if both $\mathcal{B}$ and $\neg\mathcal{B}$ were theorems of some FO predicate calculus K. Then, by Prop 2.2, both $\mathcal{B}$ and $\neg\mathcal{B}$ would be logically valid, which creates a contradiction.

If we were to allow this contradiction, we could then prove anything. For instance, the following wf is a tautology: $\mathcal{B} \Rightarrow (\neg\mathcal{B} \Rightarrow \mathcal{C})$. By MP twice, we get $\vdash \mathcal{C}$. This tells us that, in order for K to be consistent, it cannot take on every wf as a theorem.

An important note: the deduction theorem (Prop 1.9) cannot be carried over to FO theories directly. For example, for any wf $\mathcal{B}$, we can show that $\mathcal{B} \vdash_K (\forall x_i)\mathcal{B}$ however we cannot always show that $\vdash_K \mathcal{B} \Rightarrow (\forall x_i)\mathcal{B}$

Consider a domain with at least two elements, c and d . Now, let $\mathcal{B} A_1^1(x_1)$ and interpret A_1^1 as some property which holds only for c . Then A_1^1 is satisfied for any sequence $s = (s_1, s_2, \dots)$ such that $s_1 = c$, however $(\forall x_1)A_1^1(x_1)$ is not satisfied by any sequence. So, $A_1^1(x_1) \Rightarrow (\forall x_1)A_1^1(x_1)$ is not true for this interpretation, so it is not logically valid. Because it is not logically valid, by Prop 2.2 it cannot be a theorem.

We can still use the deduction theorem, we just have to be more careful to avoid this kind of mistake. Now, let $\mathcal{B}$ be a wf in a set Γ of wfs and assume that we have some sequence of wfs $\mathcal{D}_1, \dots, \mathcal{D}_n$ from Γ together with justification for each step in the deduction. We say that $\mathcal{D}_i$ depends upon $\mathcal{B}$ in this proof iff: **«Recursive definition!»**

1. $\mathcal{D}_i$ is $\mathcal{B}$ and the justification for $\mathcal{D}_i$ belongs to Γ or,
2. $\mathcal{D}_i$ is justified as a direct consequence of MP or Gen of some preceding wfs of the sequence, where at least one of the preceding wfs depends upon $\mathcal{B}$.

There is a good example on page 71 of this.

Proposition 2.4

If $\mathcal{C}$ does not depend upon $\mathcal{B}$ in a deduction showing that $\Gamma, \mathcal{B} \vdash \mathcal{C}$, then $\Gamma \vdash \mathcal{C}$.

Proof This one is pretty simple, check page 71 if needed. Basically induction on the length n of the deduction, consider the general cases and boom done.

Proposition 2.5 (The Deduction Theroem)

Assume that, in some deduction showing that $\Gamma, \mathcal{B} \vdash \mathcal{C}$, no application of Gen to a wf that depends upon $\mathcal{B}$ has as its quantified variable a free variable of $\mathcal{B}$. Then $\Gamma \vdash \mathcal{B} \Rightarrow \mathcal{C}$.

Proof From here on out, for longer and more dense proofs, I will be listing the steps for easier reading.

Let $\mathcal{D}_1, \dots, \mathcal{D}_n$ be a deduction of $\mathcal{C}$ from Γ and $\mathcal{B}$ which satisfies the assumption of the proposition. **Note that in this deduction, $\mathcal{D}_n$ is $\mathcal{C}$.** Now, we prove by induction that $\Gamma \vdash \mathcal{B} \Rightarrow \mathcal{D}_i$ for each $i \leq n$:

1. If $\mathcal{D}_i$ is an axiom or belongs to Γ , then $\Gamma \vdash \mathcal{B} \Rightarrow \mathcal{D}_i$ because $\mathcal{D}_i \Rightarrow (\mathcal{B} \Rightarrow \mathcal{D}_i)$ is an instance of (A1).

2. If $\mathcal{D}_i$ is $\mathcal{B}$, then $\Gamma \vdash \mathcal{B} \Rightarrow \mathcal{D}_i$ because $\vdash \mathcal{B} \Rightarrow \mathcal{B}$ by Prop 2.1 (Tautology).
3. If there exists a j and k less than i such that $\mathcal{D}_k$ is $\mathcal{D}_j \Rightarrow \mathcal{D}_i$, then by **IH**, $\Gamma \vdash \mathcal{B} \Rightarrow \mathcal{D}_j$ and $\Gamma \vdash \mathcal{B} \Rightarrow \mathcal{D}_k$, with the latter substituting $\mathcal{D}_k$ to get $\Gamma \vdash \mathcal{B} \Rightarrow (\mathcal{D}_j \Rightarrow \mathcal{D}_i)$.
 - a. Now, by axiom (A2), $\vdash (\mathcal{B} \Rightarrow (\mathcal{D}_j \Rightarrow \mathcal{D}_i)) \Rightarrow ((\mathcal{B} \Rightarrow \mathcal{D}_j) \Rightarrow (\mathcal{B} \Rightarrow \mathcal{D}_i))$.
 - b. Now, by MP twice, we get $\vdash \mathcal{B} \Rightarrow \mathcal{D}_i$.
4. Now, suppose that $j < i$ such that $\mathcal{D}_i$ is $(\forall x_k)\mathcal{D}_j$, so, By **IH**, $\Gamma \vdash \mathcal{B} \Rightarrow \mathcal{D}_j$.
5. By hypothesis of the theorem, either $\mathcal{D}_j$ does not depend upon $\mathcal{B}$ or x_k is not a free variable of $\mathcal{B}$.
6. If $\mathcal{D}_j$ does not depend upon $\mathcal{B}$:
 - a. By Prop 2.4, $\Gamma \vdash \mathcal{D}_j$, and then by Gen we see that $\Gamma \vdash (\forall x_k)\mathcal{D}_j$.
 - b. Because $\mathcal{D}_i$ is $(\forall x_k)\mathcal{D}_j$ as assumed in step 4, $\Gamma \vdash \mathcal{D}_i$.
 - c. Now, by (A1), $\vdash \mathcal{D}_i \Rightarrow (\mathcal{B} \Rightarrow \mathcal{D}_i)$.
 - d. By MP of step b on step c, we see that $\Gamma \vdash \mathcal{B} \Rightarrow \mathcal{D}_i$.
7. On the other hand, if x_k is not a free variable of $\mathcal{B}$:
 - a. By (A5), $\vdash (\forall x_k)(\mathcal{B} \Rightarrow \mathcal{D}_j) \Rightarrow (\mathcal{B} \Rightarrow (\forall x_k)\mathcal{D}_j)$.
 - b. By **IH**, we know that $\Gamma \vdash \mathcal{B} \Rightarrow \mathcal{D}_j$.
 - c. So, by Gen on step b, we have $\Gamma \vdash (\forall x_k)(\mathcal{B} \Rightarrow \mathcal{D}_j)$.
 - d. By MP of step c on step a, we get $\Gamma \vdash \mathcal{B} \Rightarrow (\forall x_k)\mathcal{D}_j$.
 - e. Because $\mathcal{D}_i$ is $(\forall x_k)\mathcal{D}_j$ as assumed in step 4, $\Gamma \vdash \mathcal{D}_i$.

So, we have shown by induction that $\Gamma \vdash \mathcal{B} \Rightarrow \mathcal{D}_i$ for each $i \leq n$. Our proposition is just the special case where $i = n$. This can be cumbersome, so Mendelson presents the following Corollaries which are easier to use.

Corollary 2.6 If a deduction showing that $\Gamma, \mathcal{B} \vdash \mathcal{C}$ involves no application of Gen of which the quantified variables is free in $\mathcal{B}$, then $\Gamma \vdash \mathcal{B} \Rightarrow \mathcal{C}$.

Corollary 2.7 If $\mathcal{B}$ is a closed wf and $\Gamma, \mathcal{B} \vdash \mathcal{C}$, then $\Gamma \vdash \mathcal{B} \Rightarrow \mathcal{C}$.

Useful extension of Propositions 2.4-2.7

I am a little confused about Mendelson's explanation here, but the upshot is that we can apply the deduction theorem many times in a row. E.g. $\Gamma, \mathcal{D}, \mathcal{B} \vdash \mathcal{C}$ yields $\Gamma \vdash \mathcal{D} \Rightarrow (\mathcal{B} \Rightarrow \mathcal{C})$

Example

$$\vdash (\forall x_1)(\forall x_2)\mathcal{B} \Rightarrow (\forall x_2)(\forall x_1)\mathcal{B}$$

Proof

- | | | |
|----|--|------|
| 1. | $(\forall x_1)(\forall x_2)\mathcal{B}$ | Hyp |
| 2. | $(\forall x_1)(\forall x_2)\mathcal{B} \Rightarrow (\forall x_2)\mathcal{B}$ | (A4) |

- | | | |
|----|--|----------|
| 3. | $(\forall x_2)\mathcal{B}$ | 1, 2, MP |
| 4. | $(\forall x_2)\mathcal{B} \Rightarrow \mathcal{B}$ | (A4) |
| 5. | $\mathcal{B}$ | 3, 4, MP |
| 6. | $(\forall x_1)\mathcal{B}$ | 5, Gen |
| 7. | $(\forall x_2)(\forall x_1)\mathcal{B}$ | 6, Gen |

By 1-7, we have $(\forall x_1)(\forall x_2)\mathcal{B} \vdash (\forall x_2)(\forall x_1)\mathcal{B}$ where no application of Gen has as a quantified variable a free variable of $(\forall x_1)(\forall x_2)\mathcal{B}$. So, by Corollary 2.6, $\vdash (\forall x_1)(\forall x_2)\mathcal{B} \Rightarrow (\forall x_2)(\forall x_1)\mathcal{B}$.

2.5 Additional Metatheorems and Derived Rules

2.5.1 Particularization Rule A4

If t is free for x in $\mathcal{B}(x)$, then $(\forall x)\mathcal{B}(x) \vdash \mathcal{B}(t)$.

2.5.2 Existential Rule E4

Let t be a term that is free for x in a wf $\mathcal{B}(x, t)$, and let $\mathcal{B}(t, t)$ arise from $\mathcal{B}(x, t)$ by replacing all free occurrences of x by t . ($\mathcal{B}(x, t)$ may or may not contain occurrences of t) Then, $\mathcal{B}(t, t) \vdash (\exists x)\mathcal{B}(x, t)$.

There are also derived rules for all the normal stuff from before (e.g. negation elimination, conjunction introduction, biconditional elimination, proof by contradiction, etc.). Refer to page 74 for more.

Lemma 2.8

For any wfs $\mathcal{B}$ and $\mathcal{C}$, $\vdash (\forall x)(\mathcal{B} \Leftrightarrow \mathcal{C}) \Rightarrow ((\forall x)\mathcal{B} \Leftrightarrow (\forall x)\mathcal{C})$.

Proof

- | | | |
|-----|---|-------------------------------|
| 1. | $(\forall x)(\mathcal{B} \Leftrightarrow \mathcal{C})$ | Hyp |
| 2. | $(\forall x)\mathcal{B}$ | Hyp |
| 3. | $\mathcal{B} \Leftrightarrow \mathcal{C}$ | 1, rule A4 |
| 4. | $\mathcal{B}$ | 2, rule A4 |
| 5. | $\mathcal{C}$ | 3, 4, $\Leftrightarrow$ elim |
| 6. | $(\forall x)\mathcal{C}$ | 5, Gen |
| 7. | $(\forall x)(\mathcal{B} \Leftrightarrow \mathcal{C}), (\forall x)\mathcal{B} \vdash (\forall x)\mathcal{C}$ | 1-6 |
| 8. | $(\forall x)(\mathcal{B} \Leftrightarrow \mathcal{C}) \vdash (\forall x)\mathcal{B} \Rightarrow (\forall x)\mathcal{C}$ | 1-7, Corollary 2.6 |
| 9. | $(\forall x)(\mathcal{B} \Leftrightarrow \mathcal{C}) \vdash (\forall x)\mathcal{C} \Rightarrow (\forall x)\mathcal{B}$ | Proof like 8 |
| 10. | $(\forall x)(\mathcal{B} \Leftrightarrow \mathcal{C}) \vdash (\forall x)\mathcal{B} \Leftrightarrow (\forall x)\mathcal{C}$ | 8, 9, $\Leftrightarrow$ intro |
| 11. | $\vdash (\forall x)(\mathcal{B} \Leftrightarrow \mathcal{C}) \Rightarrow ((\forall x)\mathcal{B} \Leftrightarrow (\forall x)\mathcal{C})$ | 1-10, Corollary 2.6 |

Proposition 2.9

If $\mathcal{C}$ is a subformula of $\mathcal{B}$, $\mathcal{B}'$ is the result of replacing zero or more occurrences of $\mathcal{C}$ in $\mathcal{B}$ by a wf $\mathcal{D}$, and every free variable of $\mathcal{C}$ or $\mathcal{D}$ that is also a bound variable of $\mathcal{B}$ occurs in the list $y_1, \dots, y_k$, then:

- a. $\vdash [(\forall y_1) \dots (\forall y_k)(\mathcal{C} \Leftrightarrow \mathcal{D})] \Rightarrow (\mathcal{B} \Leftrightarrow \mathcal{B}')$ (Equivalence theorem)
- b. If $\vdash \mathcal{C} \Leftrightarrow \mathcal{D}$, then $\vdash \mathcal{B} \Leftrightarrow \mathcal{B}'$ (Replacement theorem)
- c. If $\vdash \mathcal{C} \Leftrightarrow \mathcal{D}$ and $\vdash \mathcal{B}$, then $\vdash \mathcal{B}'$

Example

$$\vdash (\forall x)(A_1^1(x) \Leftrightarrow A_2^1(x)) \Rightarrow [(\exists x)A_1^1(x) \Leftrightarrow (\exists x)A_2^1(x)]$$

Proof We use induction on the number of connectives and quantifiers in $\mathcal{B}$.

- 1. If zero occurrences are replaced, $\mathcal{B}'$ is $\mathcal{B}$ and the wf to be proved is an instance of the tautology $A \Rightarrow (B \Leftrightarrow B)$.
- 2. If $\mathcal{C}$ is identical to $\mathcal{B}$ and this occurrence of $\mathcal{C}$ is replaced by $\mathcal{D}$, the wf

$$[(\forall y_1) \dots (\forall y_k)(\mathcal{C} \Leftrightarrow \mathcal{D})] \Rightarrow (\mathcal{B} \Rightarrow \mathcal{B}')$$

is derivable by Exercise 2.27(d). Here is the theorem from 2.27(d):

$$\vdash (\forall y_1) \dots (\forall y_k)\mathcal{B} \Rightarrow \mathcal{B}$$

I might figure out how to prove this later idk. This is the base case.

- 3. So, we can assume that $\mathcal{C}$ is a proper part of $\mathcal{B}$ (e.g. not identical, think proper subset versus subset) and that at least one occurrence of $\mathcal{C}$ is replaced.
- 4. **IH:** the result holds for all wfs with fewer connectives and quantifiers than $\mathcal{B}$. Now we consider all cases which break down $\mathcal{B}$ into smaller parts for our inductive step.

Case 1. $\mathcal{B}$ is an atomic wf. So, $\mathcal{C}$ cannot be a proper part of $\mathcal{B}$.

Case 2. $\mathcal{B}$ is $\neg\mathcal{E}$. Let $\mathcal{B}'$ be $\neg\mathcal{E}'$. By **IH**:

$$\vdash [(\forall y_1) \dots (\forall y_k)(\mathcal{C} \Leftrightarrow \mathcal{D})] \Rightarrow (\mathcal{E} \Leftrightarrow \mathcal{E}')$$

Now we find a suitable tautology:

$$(C \Rightarrow (A \Leftrightarrow B)) \Rightarrow (C \Rightarrow (\neg A \Leftrightarrow \neg B))$$

Let $[(\forall y_1) \dots (\forall y_k)(\mathcal{C} \Leftrightarrow \mathcal{D})]$ be C , let $\mathcal{E}$ be A , and $\mathcal{E}'$ be B . Notice that after this substitution, the premise of our tautology is of the same form as our **IH**. So, by MP, we obtain:

$$C \Rightarrow (\neg A \Leftrightarrow \neg B)$$

And, if we substitute our wfs back in we get:

$$\vdash [(\forall y_1) \dots (\forall y_k)(\mathcal{C} \Leftrightarrow \mathcal{D})] \Rightarrow (\neg\mathcal{E} \Leftrightarrow \neg\mathcal{E}')$$

Recall that $\mathcal{B}$ is $\neg\mathcal{E}$ and $\mathcal{B}'$ is $\neg\mathcal{E}'$. So, we obtain:

$$\vdash [(\forall y_1) \dots (\forall y_k)(\mathcal{C} \Leftrightarrow \mathcal{D})] \Rightarrow (\mathcal{B} \Leftrightarrow \mathcal{B}')$$

Case 3. $\mathcal{B}$ is $\mathcal{C} \Rightarrow \mathcal{F}$. Let $\mathcal{B}'$ be $\mathcal{C}' \Rightarrow \mathcal{F}$. By IH:

$$\vdash [(\forall y_1) \dots (\forall y_k)(\mathcal{C} \Leftrightarrow \mathcal{D})] \Rightarrow (\mathcal{C} \Leftrightarrow \mathcal{C}')$$

$$\vdash [(\forall y_1) \dots (\forall y_k)(\mathcal{C} \Leftrightarrow \mathcal{D})] \Rightarrow (\mathcal{F} \Leftrightarrow \mathcal{F}')$$

Because $\mathcal{C}$ and $\mathcal{F}$ are smaller than $\mathcal{B}$. Now we find a suitable tautology:

$$(A \Rightarrow (\mathcal{B} \Leftrightarrow C)) \wedge (A \Rightarrow (D \Leftrightarrow E)) \Rightarrow (A \Rightarrow [(B \Rightarrow D) \Leftrightarrow (C \Rightarrow E)])$$

This might seem like a scarily long and arbitrary wf. However, if you follow along closely, you can see that it actually works out very nicely. We follow a similar procedure to case 2. Let $[(\forall y_1) \dots (\forall y_k)(\mathcal{C} \Leftrightarrow \mathcal{D})]$ be A , $\mathcal{C}$ be B , and so on. Notice that the premise of our tautology matches both applications of **IH** joined by $\wedge$. So, by MP we obtain the antecedent of our tautology. Substitute back in our wfs and we obtain:

$$\vdash [(\forall y_1) \dots (\forall y_k)(\mathcal{C} \Leftrightarrow \mathcal{D})] \Rightarrow (\neg \mathcal{C} \Leftrightarrow \neg \mathcal{C}')$$

Case 4. $\mathcal{B}$ is $(\forall x)\mathcal{C}$. Let $\mathcal{B}'$ be $(\forall x)\mathcal{C}'$. By **IH**:

$$\vdash [(\forall y_1) \dots (\forall y_k)(\mathcal{C} \Leftrightarrow \mathcal{D})] \Rightarrow (\mathcal{C} \Leftrightarrow \mathcal{C}')$$

Now, x does not occur free in $(\forall y_1) \dots (\forall y_k)(\mathcal{C} \Leftrightarrow \mathcal{D})$ because, if it did, it would be free in $\mathcal{C}$ or $\mathcal{D}$ and, since it is bound in $\mathcal{B}$, it would be one of $y_1, \dots, y_k$. and it would not be free in $(\forall y_1) \dots (\forall y_k)(\mathcal{C} \Leftrightarrow \mathcal{D})$. Hence, by axiom (A5) we obtain:

$$\vdash (\forall y_1) \dots (\forall y_k)(\mathcal{C} \Leftrightarrow \mathcal{D}) \Rightarrow (\forall x)(\mathcal{C} \Leftrightarrow \mathcal{C}')$$

By Lemma 2.8, we also obtain: **«Typo on this statement on pg 77!»**

$$\vdash (\forall x)(\mathcal{C} \Leftrightarrow \mathcal{C}') \Rightarrow ((\forall x)\mathcal{C} \Leftrightarrow (\forall x)\mathcal{C}')$$

Then, by a suitable tautology and MP:

$$\vdash [(\forall y_1) \dots (\forall y_k)(\mathcal{C} \Leftrightarrow \mathcal{D})] \Rightarrow (\mathcal{B} \Leftrightarrow \mathcal{B}')$$

5. From $\vdash \mathcal{C} \Leftrightarrow \mathcal{D}$, by several (in particular, k) applications of Gen, we obtain $\vdash (\forall y_1) \dots (\forall y_k)(\mathcal{C} \Leftrightarrow \mathcal{D})$. Then, by step 4 and MP, $\vdash \mathcal{B} \Leftrightarrow \mathcal{B}'$.

6. Use step 5 and $\Leftrightarrow$ elimination.

«Conceptual Note»

Notice that having as few inference rules as possible makes for fewer steps during induction. The proof for Proposition 2.9 was already quite verbose; imagine if we had more inference rules! The nice thing about our system is that we can use other connectives (e.g. $\vee$ or $\Leftrightarrow$) in our wfs without having to prove them during induction because they can all be rewritten in terms of $\Rightarrow$ and $\neg$. Notice that we had to prove the above claim for $\mathcal{B}$ in terms of some wf of the form $\neg \mathcal{C}$ or $\mathcal{D} \Rightarrow \mathcal{F}$!

2.6 Rule C

If we have some wf of the form $(\exists x)\mathcal{B}(x)$, we can choose some b such that $\mathcal{B}(b)$. This can often simplify proofs, as this is basically a shortcut for a longer sequence of steps. Check page 78-79 for an example. There are some rules, however. A rule C deduction in a FOT K is defined in the following manner:

$\Gamma \vdash_C \mathcal{B}$ iff there is a sequence of wfs $\mathcal{D}_1, \dots, \mathcal{D}_n$ such that $\mathcal{D}_n$ is $\mathcal{B}$ and the following conditions hold:

1. For each $i < n$, either
 - a. $\mathcal{D}_i$ is an axiom of K
 - b. $\mathcal{D}_i$ is in Γ
 - c. $\mathcal{D}_i$ follows by MP or Gen from preceding wfs in the sequence
 - d. There is a preceding wf $(\exists x)\mathcal{C}(x)$ such that $\mathcal{D}_i$ is $\mathcal{C}(d)$, where d is a **new** individual constant (Rule C).
2. As axioms in condition 1a, we also can use all logical axioms that involve the new individual constants already introduced in the sequence by rule C.
3. No application of Gen is made using a variable that is free in some $(\exists x)\mathcal{C}(x)$ to which rule C has been previously applied.
4. $\mathcal{B}$ contains none of the new individual constants introduced in the sequence in any application of rule C.

Condition 3 is important because if an application of rule C to a wf $(\exists x)\mathcal{C}(x)$ yields $\mathcal{C}(d)$, then the object referred to by d may depend on the values of the free variables in $(\exists x)\mathcal{C}(x)$. So that one object may not satisfy $\mathcal{C}(x)$ for *all* values of the free variables in $(\exists x)\mathcal{C}(x)$. There is a good example on page 80 which compares the wfs $(\forall x)(\exists y)A_1^2(x, y)$ and $(\exists y)(\forall x)A_1^2(x, y)$, where there is some interpretation which makes the former true but the latter false.

Proposition 2.10 If $\Gamma \vdash_C \mathcal{B}$, then $\Gamma \vdash \mathcal{B}$. Moreover, from the following proof one can verify that, if there is an application of Gen in the new proof of $\mathcal{B}$ using Γ using a certain variable and applied to a wf depending upon a certain wf of Γ , then there was such an application of Gen in the original proof.

1. Let $(\exists y_1)\mathcal{C}_1(y_1), \dots, (\exists y_k)\mathcal{C}_k(y_k)$ be the wfs in order of occurrence to which rule C is applied in the proof of $\Gamma \vdash_C \mathcal{B}$, and let $d_1, \dots, d_k$ be the corresponding new individual constants.
2. Then, $\Gamma, \mathcal{C}_1(d_1), \dots, \mathcal{C}_k(d_k) \vdash \mathcal{B}$.
3. Now, by condition 3 of the definition above, Corollary 2.6 is now applicable, yielding $\Gamma, \mathcal{C}_1(d_1), \dots, \mathcal{C}_{k-1}(d_{k-1}) \vdash \mathcal{C}_k(d_k) \Rightarrow \mathcal{B}$.
4. We replace d_k everywhere by a variable z that does not occur in the proof.
5. So:

$$\Gamma, \mathcal{C}_1(d_1), \dots, \mathcal{C}_{k-1}(d_{k-1}) \vdash \mathcal{C}_k(z) \Rightarrow \mathcal{B}$$

6. Now, by Gen:

$$\Gamma, \mathcal{C}_1(d_1), \dots, \mathcal{C}_{k-1}(d_{k-1}) \vdash (\forall x)(\mathcal{C}_k(z) \Rightarrow \mathcal{B})$$

7. So, by Exercise 2.32(d):

$$\Gamma, \mathcal{C}_1(d_1), \dots, \mathcal{C}_{k-1}(d_{k-1}) \vdash (\exists y_k)\mathcal{C}_k(y_k) \Rightarrow \mathcal{B}$$

8. Notice that we can obtain:

$$\Gamma, \mathcal{C}_1(d_1), \dots, \mathcal{C}_{k-1}(d_{k-1}) \vdash (\exists y_k)\mathcal{C}_k(y_k)$$

9. So, by MP we obtain:

$$\Gamma, \mathcal{C}_1(d_1), \dots, \mathcal{C}_{k-1}(d_{k-1}) \vdash \mathcal{B}$$

This argument can be repeated to eliminate $\mathcal{C}_{k-1}(d_{k-1}), \dots, \mathcal{C}_1(d_1)$ to obtain $\Gamma \vdash \mathcal{B}$. I don't know why Mendelson didn't just use regular induction for this, this is a strange proof relative to the others in this chapter.

2.7 Completeness Theorems

Now we will show that the theorems of a first-order predicate calculus K are precisely the same as the logically valid wfs of K. Half of this result was proved in Proposition 2.2, but here we will establish this with more general propositions later.

A few quick definitions:

Example «Missing paren and extra bracket here in textbook»

$$(\forall x_3)[A_1^2(x_1, x_3) \vee A_1^1(x_1)] \text{ and } (\forall x_3)[A_1^2(x_2, x_3) \vee A_1^1(x_2)] \text{ are similar.}$$

Lemma 2.11 If $\mathcal{B}(x_i)$ and $\mathcal{B}(x_j)$ are similar, then $\vdash (\forall x_i)\mathcal{B}(x_i) \Leftrightarrow (\forall x_j)\mathcal{B}(x_j)$.

Proof

1. By axiom (A4), we get:

$$\vdash (\forall x_i)\mathcal{B}(x_i) \Rightarrow \mathcal{B}(x_j)$$

2. Then, be Gen we obtain:

$$\vdash (\forall x_j)((\forall x_i)\mathcal{B}(x_i) \Rightarrow \mathcal{B}(x_j))$$

3. Then, by axiom (A5) and MP, we obtain:

$$\vdash (\forall x_i)\mathcal{B}(x_i) \Rightarrow (\forall x_j)\mathcal{B}(x_j)$$

4. Now do the same thing to achieve the other direction and by $\Leftrightarrow$ introduction we get:

$$\vdash (\forall x_i)\mathcal{B}(x_i) \Leftrightarrow (\forall x_j)\mathcal{B}(x_j)$$

Lemma 2.12 If a closed wf $\neg \mathcal{B}$ of a theory K is not provable in K, and if K' is the theory obtained from K by adding $\mathcal{B}$ as a new axiom, then K' is consistent.

Proof Assume that K' is inconsistent.

1. Then, for some wf $\mathcal{C}$, $\vdash_{K'} \mathcal{C}$ and $\vdash_{K'} \neg \mathcal{C}$.
2. Now, by proposition 2.1 we can instantiate the following tautology:

$$\vdash_{K'} \mathcal{C} \Rightarrow (\neg \mathcal{C} \Rightarrow \neg \mathcal{B})$$

3. So, by MP twice, we get $\vdash_{K'} \neg \mathcal{B}$.
4. Now, any use of $\mathcal{B}$ as an axiom in a proof K' can be regarded as a hypothesis in a proof in K .
5. Hence, $\mathcal{B} \vdash_K \neg \mathcal{B}$.
6. Because $\mathcal{B}$ is closed, by corollary 2.7 we get $\vdash_K \mathcal{B} \Rightarrow \neg \mathcal{B}$
7. However, we can instantiate the following tautology by Proposition 2.1:

$$\vdash_K (\mathcal{B} \Rightarrow \neg \mathcal{B}) \Rightarrow \neg \mathcal{B}$$

8. So, by MP we obtain $\vdash_K \neg \mathcal{B}$

Step 8 contradicts the assumption that $\neg \mathcal{B}$ is not provable in K . Thus, K' must be consistent. This can be done without loss of generality: suppose that $\mathcal{B}$ were not provable in K . Similar steps show that adding $\neg \mathcal{B}$ as an axiom to K' preserve consistency.

Lemma 2.13

The set of expressions of a language $\mathcal{L}$ is denumerable. Hence, the same is true of the set of terms, the set of wfs, and the set of closed wfs.

Proof This one is assigning prime factorizations to expressions to show that the set of expressions of $\mathcal{L}$ is denumerable. We can effectively decide whether any given symbol is a symbol of $\mathcal{L}$ and we can effectively decide whether some number is the number of an expression of $\mathcal{L}$ (the same for terms, wfs, closed wfs, etc.).

Definitions

- i. A theory K is *complete* if, for every closed wf $\mathcal{B}$ of K , either $\vdash_K \mathcal{B}$ or $\vdash_K \neg \mathcal{B}$.
- ii. A theory K' is an *extensions* of a theory K if every theorem of K is a theorem of K' . We also call K a subtheory of K' in this case

These definitions will be important for the upcoming completeness theorems.

Proposition 2.14 (Lindenbaum's Lemma)

If K is a consistent theory, then there is a consistent, complete extension of K .

Proof

1. Let $\mathcal{B}_1, \mathcal{B}_2, \dots$ be an enumeration of all closed wfs of the language of K , by Lemma 2.13.
2. Define a sequence $J_0, J_1, J_2, \dots$ of theories in the following way:
 - i. J_0 is K .
 - ii. Assume that J_n is defined, with $n \geq 0$.

- iii. If it is not the case that $\vdash_{J_n} \neg \mathcal{B}_{n+1}$, then let J_{n+1} be obtained by adding $\mathcal{B}_{n+1}$ as an additional axiom.
- iv. On the other hand, if $\vdash_{J_n} \neg \mathcal{B}_{n+1}$, let $J_{n+1} = J_n$.
- 3. Let J be the theory obtained by taking as axioms all the axioms of all the J_i s, including $J_0 = K$.
- 4. We can show that J is consistent by proving that every J_i is consistent via a proof by contradiction in J , involving as it does only a finite number of axioms, is the same as a proof by contradiction in some J_i .
- 5. We proceed with proof by induction:
 - i. By hypothesis, $J_0 = K$ is consistent.
 - ii. Assume that J_i is consistent.
 - iii. If $J_{i+1} = J_i$, then J_{i+1} is consistent.
 - iv. If $J_{i+1} \neq J_i$, we know by definition of J_{i+1} in line 2.iii that $\neg \mathcal{B}_{i+1}$ is not provable in J_i .
 - v. This means that $\mathcal{B}_{i+1}$ is added as an axiom to J_i to generate J_{i+1} , which is consistent by Lemma 2.12.
- 6. So, all J_i s are consistent.
- 7. To prove the completeness of J , let $\mathcal{C}$ be any closed wf of K .
- 8. Then $\mathcal{C} = \mathcal{B}_{k+1}$ for some $k \geq 0$.
- 9. Now, either $\vdash_{J_k} \neg \mathcal{B}_{k+1}$ or $\vdash_{J_{k+1}} \mathcal{B}_{k+1}$
- 10. This is because if it is not the case that $\vdash_{J_k} \neg \mathcal{B}_{k+1}$, then $\mathcal{B}_{k+1}$ is added as an axiom in J_{k+1} .
- 11. Therefore, either $\vdash_J \neg \mathcal{B}_{k+1}$ or $\vdash_J \mathcal{B}_{k+1}$, which shows that J is complete.

Note that even if one can effectively determine whether any wf is an axiom of K , it may not be possible to do the same with (or even to enumerate effectively) the axioms of J ; that is, J may not be axiomatic even if K is. This is due to the possibility of not being able to determine, at each step, whether or not $\neg \mathcal{B}_{n+1}$ is provable in J_n .

Lemma 2.15 Every consistent theory K has a consistent extension K' such that K' is a scapegoat theory and K' contains denumerably many closed terms. This proof is horrific to read in the textbook, hopefully this presentation is helpful.

Proof

- 1. Add to the symbols of K a denumerable set $\{b_1, b_2, \dots\}$ of new individual constants and call this new theory K_0 .
- 2. Its axioms are those of K plus those logical axioms that involve the symbols of K and the new constants.
- 3. K_0 is consistent. If this were not the case, then there is a proof in K_0 of $\mathcal{B} \wedge \neg \mathcal{B}$.
- 4. Replace each b_i appearing in this proof by a variable that does not appear in the proof.

5. This transforms the axioms into axioms and preserves the correctness of the applications of the rules of inference.
6. The final wf in the proof is still a contradiction, but now the proof does not involve b_i s and is therefore a proof in K .
7. This contradicts the consistency of K , hence K_0 is consistent.
8. By Lemma 2.13, let $F_1(x_{i_1}), F_2(x_{i_2}), \dots, F_k(x_{i_k}), \dots$ be an enumeration of all wfs of K_0 that have one free variable.
9. Choose a sequence $b_{j_1}, b_{j_2}, \dots$ of some of the new individual constants such that each b_{j_k} is not contained in any of the wfs $F_1(x_{i_1}), \dots, F_k(x_{i_k})$ and such that b_{j_k} is distinct from each of $b_{j_1}, \dots, b_{j_{k-1}}$.
10. Consider the wf:

$$(S_k) \quad (\exists x_{i_k}) \neg F_k(x_{i_k}) \Rightarrow \neg F_k(b_{j_k})$$

11. Let K_n be the theory obtained by adding $(S_1), \dots, (S_n)$ to the axioms of K_0 , and let K_∞ be the theory obtained by adding all the (S_i) s as axioms to K_0 .
12. Any proof in K_∞ contains only a finite number of the (S_i) s and, therefore, will also be a proof in some K_n .
13. Hence, if all of the K_n s are consistent, so is K_∞ .
14. To demonstrate all K_n s are consistent, we proceed by induction:
 - i. We know that K_0 is consistent.
 - ii. Assume that K_{n-1} is consistent but that K_n is inconsistent ($n \geq 1$).
 - iii. If K_n is inconsistent, then any wf is provable in K_n by the tautology $\neg A \Rightarrow (A \Rightarrow B)$ by Proposition 2.1 and MP. In particular, $\vdash_{K_n} \neg(S_n)$.
 - iv. Hence, $(S_n) \vdash_{K_{n-1}} \neg(S_n)$.
 - v. Because (S_n) is closed, we can use Corollary 2.7 to obtain $\vdash_{K_{n-1}} (S_n) \Rightarrow \neg(S_n)$.
 - vi. Now, we can use the tautology $(A \Rightarrow \neg A) \Rightarrow \neg A$ by Proposition 2.1 and MP to obtain $\vdash_{K_{n-1}} \neg(S_n)$. That is,

$$\vdash_{K_{n-1}} \neg[(\exists x_{i_n}) \neg F_n(x_{i_n}) \Rightarrow \neg F_n(b_{j_n})]$$

- vii. Now, by $\Rightarrow$ elimination, we obtain $\vdash_{K_{n-1}} (\exists x_{i_n}) \neg F_n(x_{i_n})$ and $\vdash_{K_{n-1}} \neg \neg F_n(b_{j_n})$, and then, by $\neg$ elimination, $\vdash_{K_{n-1}} F_n(b_{j_n})$.
- viii. From the latter and the fact that b_{j_n} does not occur in $(S_0), \dots, (S_{n-1})$, we conclude that $\vdash_{K_{n-1}} F_n(x_r)$, where x_r is a variable that does not occur in the proof of $F_n(b_{j_n})$. (Simply replace in the proof all occurrences of b_{j_n} by x_r).
- ix. By Gen, $\vdash_{K_{n-1}} (\forall x_r) F_n(x_r)$, and then, by Lemma 2.11 and $\Leftrightarrow$ elimination, $\vdash_{K_{n-1}} (\forall x_{i_n}) F_n(x_{i_n})$. (We use the fact that $F_n(x_r)$ and $F_n(x_{i_n})$ are similar).
- x. But we already have $\vdash_{K_{n-1}} (\exists x_{i_n}) \neg F_n(x_{i_n})$, which is an abbreviation of $\vdash_{K_{n-1}} \neg(\forall x_{i_n}) \neg \neg F_n(x_{i_n})$.

- xi. This yields $\vdash_{K_{n-1}} \neg(\forall x_{i_n}) F_n(x_{i_n})$ by replacement theorem, which contradicts the hypothesis that K_{n-1} is consistent.
- 15. Hence, K_n must also be consistent.
- 16. Thus, K_∞ is consistent, it is an extension of K , and it is a scapegoat theory.

Lemma 2.16 Let J be a consistent, complete scapegoat theory. Then J has a model M whose domain is the set D of closed terms of J .

Proof

- 1. For any individual constant a_i of J , let $(a_i)^M = a_i$.
- 2. For any function letter f_k^n of J and for any closed terms $t_1, \dots, t_n$ of J , let $(f_k^n)^M(t_1, \dots, t_n) = f_k^n(t_1, \dots, t_n)$. (Notice that $f_k^n(t_1, \dots, t_n)$ is a closed term, hence $(f_k^n)^M$ is an n -ary operation on D).
- 3. For any predicate letter A_k^n of J , let $(A_k^n)^M$ consist of all n -tuples $\langle t_1, \dots, t_n \rangle$ of closed terms $t_1, \dots, t_n$ of J such that $\vdash_J A_k^n(t_1, \dots, t_n)$.
- 4. We will now show that, for any closed wf $\mathcal{C}$ of J

$$(\Box) \quad \models_M \mathcal{C} \text{ iff } \vdash_J \mathcal{C}$$

- 5. If this is established and $\mathcal{B}$ is any axiom of J , let $\mathcal{C}$ be the closure of $\mathcal{B}$. Then:
 - i. By Gen, $\vdash_J \mathcal{C}$.
 - ii. By $(\Box)$, $\models_M \mathcal{C}$.
 - iii. By VI on page 5 ($\models_M \mathcal{B}$ iff $\models_M (\forall x_i) \mathcal{B}$), $\models_M \mathcal{B}$.
 - iv. Hence, M would be a model of J .
- 6. We now prove $(\Box)$ by induction on the number r of connectives and quantifiers in $\mathcal{C}$.
- 7. Assume that $(\Box)$ holds for all closed wfs with fewer than r connectives and quantifiers.

Case 1. $\mathcal{C}$ is a closed atomic wf $A_k^n(t_1, \dots, t_n)$. Then $(\Box)$ is a direct consequence of the definition of $(A_k^n)^M$.

Case 2. $\mathcal{C}$ is $\neg \mathcal{D}$.

- i. If $\mathcal{C}$ is true for M , then $\mathcal{D}$ is false for M .
- ii. So, by **IH**, $\text{not-}\vdash_J \mathcal{D}$.
- iii. Because J is complete and $\mathcal{D}$ is closed, $\vdash_J \neg \mathcal{D}$ «conceptual check: why can we say this?», which is the same as saying $\vdash_J \mathcal{C}$.
- iv. Conversely, if $\mathcal{C}$ is not true for M , then $\mathcal{D}$ is true for M .
- v. Hence, $\vdash_J \mathcal{D}$.
- vi. Because J is consistent, $\text{not-}\vdash_J \neg \mathcal{D}$, so $\text{not-}\vdash_J \mathcal{C}$.

Case 3. $\mathcal{C}$ is $\mathcal{D} \Rightarrow \mathcal{E}$.

- i. Since $\mathcal{C}$ is closed, so are $\mathcal{D}$ and $\mathcal{E}$.

ii. If $\mathcal{C}$ is false for M, then $\mathcal{D}$ is true and $\mathcal{E}$ is false.

iii. Hence, by **IH**, $\vdash_J \mathcal{D}$ and $\text{not-}\vdash_J \mathcal{E}$.

iv. By the completeness of J, $\vdash_J \neg \mathcal{E}$.

v. Now by instance of the following tautology:

$$D \Rightarrow (\neg E \Rightarrow \neg(D \Rightarrow E))$$

and two applications of MP, $\vdash_J \neg(\mathcal{D} \Rightarrow \mathcal{E})$.

vi. This tells us that $\vdash_J \neg \mathcal{C}$, and by consistency of J, $\text{not-}\vdash_J \mathcal{C}$.

vii. Conversely, if $\text{not-}\vdash_J \mathcal{C}$, then, by completeness of J, $\vdash_J \neg \mathcal{C}$.

viii. This is the same as $\vdash_J \neg(\mathcal{D} \Rightarrow \mathcal{E})$.

ix. By $\Rightarrow$ elimination, $\vdash_J \mathcal{D}$ and $\vdash_J \neg \mathcal{E}$.

x. Now, by $(\Box)$ for $\mathcal{D}$, $\mathcal{D}$ is true for M.

xi. By the consistency of J, $\text{not-}\vdash_J \mathcal{E}$ and, therefore, by $(\Box)$ for $\mathcal{E}$, $\mathcal{E}$ is false for M.

xii. Thus, since $\mathcal{D}$ is true for M and $\mathcal{E}$ is false for M, $\mathcal{C}$ is false for M.

Case 4. $\mathcal{C}$ is $(\forall x_m)\mathcal{D}$.

Case 4a. $\mathcal{D}$ is a closed wf.

i. By **IH**, $\models_M \mathcal{D}$ iff $\vdash_J \mathcal{D}$.

ii. By Exercise 2.32(a)¹, $\vdash_J \mathcal{D} \Leftrightarrow (\forall x_m)\mathcal{D}$.

iii. So, $\vdash_J \mathcal{D}$ iff $\vdash_J (\forall x_m)\mathcal{D}$ by $\Leftrightarrow$ elimination.

iv. Also, $\models_M \mathcal{D}$ iff $\models_M (\forall x_m)\mathcal{D}$ by property VI on page 5.

v. Hence, $\models_M \mathcal{C}$ iff $\models_M (\forall x_m)\mathcal{C}$.

Case 4b. $\mathcal{D}$ is not a closed wf. Since $\mathcal{C}$ is closed, $\mathcal{D}$ has x_m as its only free variable. Let $\mathcal{D}$ be $F(x_m)$. Then $\mathcal{C}$ is $(\forall x_m)F(x_m)$

i. Assume $\models_M \mathcal{C}$ and $\text{not-}\vdash_J \mathcal{C}$.

a. By the completeness of J, $\vdash_J \neg \mathcal{C}$, which means $\vdash_J \neg(\forall x_m)F(x_m)$.

b. Then, by Exercise 2.33(a)² and $\Leftrightarrow$ elimination, $\vdash_J$.

c. Because J is a scapegoat theory, $\vdash_J \neg F(t)$ for some closed term t of J.

d. By $\models_M \mathcal{C}$, which means $\models_M (\forall x_m)F(x_m)$.

e. Because $(\forall x_m)F(x_m) \Rightarrow F(t)$ is true for M by property X on page 6, $\models_M F(t)$.

f. Hence, by $(\Box)$ for $F(t)$, $\vdash_J F(t)$.

g. This contradicts the consistency of J.

¹Assume that $\mathcal{B}$ and $\mathcal{C}$ are wfs and that x is not free in $\mathcal{B}$. Prove: $\vdash \mathcal{B} \Rightarrow (\forall x)\mathcal{B}$.

²Prove that $\vdash (\exists x)\neg \mathcal{B} \Leftrightarrow \neg(\forall x)\mathcal{B}$

- h. So, if $\models_M \mathcal{C}$, then $\vdash_J \mathcal{C}$.
- ii. Assume $\text{not-}\models_M \mathcal{C}$ and $\vdash_J \mathcal{C}$. Then:
- $$(\#) \quad \vdash_J (\forall x_m)F(x_m) \quad \text{and} \quad \text{not-}\models_M (\forall x_m)F(x_m)$$
- By $(\#\#)$, some sequence of elements of the domain D does not satisfy $(\forall x_m)f(x_m)$.
 - Hence, some sequence s does not satisfy $F(x_m)$. Let t be the i th component of s .
 - Notice that $s^*(u) = u$ for all closed terms u of J (by the definition of $(a_i)^M$ and $(f_k^n)^M$).
 - It can also be seen that $F(t)$ has fewer connectives and quantifiers than $\mathcal{C}$.
 - Therefore, the **IH** applies to $F(t)$, e.g. $(\Box)$ holds for $F(t)$.
 - Hence, by Lemma 2(a) on page 60 of the textbook (sorry I didn't include this one), s does not satisfy $F(t)$.
 - So, $F(t)$ is false for M .
 - However, by $(\#)$ and rule A4, $\vdash_J F(t)$, and so, by $(\Box)$ for $F(t)$, $\models_M F(t)$.
 - This contradiction shows that, if $\vdash_J \mathcal{C}$, then $\models_M \mathcal{C}$.

Proposition 2.17 Every consistent theory K has a denumerable model.

Proof

- By Lemma 2.15, K has a consistent extension K' such that K' is a scapegoat theory and has denumerably many closed terms.
- By Lindenbaum's lemma (Proposition 2.14), K' has a consistent, complete extension J that has the same symbols as K' .
- Hence, J is also a scapegoat theory.
- By Lemma 2.16, J has a model M whose domain is the denumerable set of closed terms of J .
- Since J is an extension of K , M is a denumerable model of K .

Corollary 2.18 Any logically valid wf of $\mathcal{B}$ of a theory K is a theorem of K .

Proof

- We need consider only closed wfs $\mathcal{B}$, since a wf $\mathcal{D}$ is logically valid iff its closure is logically valid and $\mathcal{D}$ is provable in K iff its closure is provable in K .
- So, let $\mathcal{B}$ be a logically valid closed wf of K . Assume that $\text{not-}\vdash_K \mathcal{B}$.
- By Lemma 2.12, if we add $\neg\mathcal{B}$ as a new axiom to K , the new theory K' is consistent.
- Hence, by proposition 2.17, K' has a model M .
- Since $\neg\mathcal{B}$ is an axiom of K , $\neg\mathcal{B}$ is true for M .
- But, since $\mathcal{B}$ is logically valid, $\mathcal{B}$ is true for M .
- Hence, $\mathcal{B}$ is both true and false for M , which is impossible.
- Thus, $\mathcal{B}$ must be a theorem of K .

Corollary 2.19 Gödel's Completeness Theorem, 1930) In any predicate calculus, the theorems are precisely the logically valid wfs.

Proof This follows from Proposition 2.2 and Corollary 2.18. That's all Mendelson gives us, but really it makes sense: 2.2 handles all the wfs of our predicate calculus which are "imported" from our chapter 1 propositional calculus (i.e. no quantifiers, etc.), while 2.18 handles all other wfs, e.g. the wfs which are unique to our predicate calculus.

Corollary 2.20 Let K be any theory.

1. A wf $\mathcal{B}$ is true in every denumerable model of K iff $\vdash_K \mathcal{B}$.
2. If (in every model of K) every sequence that satisfies all wfs in a set Γ of wfs also satisfies a wf $\mathcal{B}$, then $\Gamma \vdash_K \mathcal{B}$.
3. If a wf $\mathcal{B}$ is a logical consequence of a set Γ of wfs of K , then $\Gamma \vdash_K \mathcal{B}$.
4. If a wf $\mathcal{B}$ of K is a logical consequence of a wf $\mathcal{C}$ of K , then $\mathcal{C} \vdash_K \mathcal{B}$.

Proof

- a. We may assume $\mathcal{B}$ is closed (Why?)³.
 - i. If $\text{not-}\vdash_K \mathcal{B}$, then the theory $K' = K + \{\neg\mathcal{B}\}$ is consistent, by lemma 2.12 (the notation indicates that $\mathcal{B}$ is added to the set of axioms of K).
 - ii. Hence, by Proposition 2.17, K' has a denumerable model M .
 - iii. However, because $\neg\mathcal{B}$ is an axiom of K' , it is true for M .
 - iv. By hypothesis, since M is a denumerable model of K , $\mathcal{B}$ is true for M .
 - v. This would indicate that $\mathcal{B}$ is true and false for M , which is impossible.
- b. Consider the theory $K + \Gamma$. By the hypothesis, $\mathcal{B}$ is true for every model of this theory. Hence, by (a), $\vdash_{K+\Gamma} \mathcal{B}$. So, $\Gamma \vdash_K \mathcal{B}$.
- c. Part (c) is a consequence of (b) and part (d) is a special case of (c).

Corollary 2.21 (Skolem-Löwenheim Theorem, 1920, 1915) Any theory that has a model has a denumerable model.

Proof If K has a model, then K is consistent, since no wf can be both true and false for the same model M . Hence, by proposition 2.17, K has a denumerable model.

2.8 First-Order Theories with Equality

In this chapter we introduce the standard interpretation, i.e. a first-order theory with equality. Let A_1^2 be a predicate letter for which $A_1^2(t, s)$ indicates that $t = s$, and vice versa $\neg A_1^2(t, s)$ indicates that $t \neq s$. More specifically, K is a first-order theory of equality if the followin are theorems of K :

(A6) $(\forall x_1)x_1 = x_1$ (reflexivity principle)

(A7) $x = y \Rightarrow (\mathcal{B}(x, x) \Rightarrow \mathcal{B}(x, y))$ (substitutivity of equality)

³ Open wfs cannot be true in every model! (fact check here would be nice but I'm pretty sure this is true)

Where x and y are any variables, $\mathcal{B}(x, y)$ is any wf, and $\mathcal{B}(x, y)$ arises from $\mathcal{B}(x, x)$ by replacing some (but not necessarily all) free occurrences of x by y provided that y is free for x in $\mathcal{B}(x, x)$. So, $\mathcal{B}(x, y)$ may or may not contain free occurrences of x . Note that the numbering of the above axioms is a continuation of the logical axioms (A1)-(A5).

Proposition 2.23

In any theory of equality,

1. $\vdash t = t$ for any term t .
2. $\vdash t = s \Rightarrow s = t$ for any terms t and s .
3. $\vdash t = s \Rightarrow (s = r \Rightarrow t = r)$ for any terms t, s , and r .

Proof

- a. By (A6), $\vdash (\forall x_1)x_1 = x_1$. Hence, by rule A4, $\vdash t = t$.
- b. i. Let x and y be variables not occurring in t or s .
ii. Letting $\mathcal{B}(x, x)$ be $x = x$ and $\mathcal{B}(x, y)$ be $x = y$ in schema (A7):

$$\vdash x = y \Rightarrow (x = x \Rightarrow x = y)$$

iii. But, by (a), $\vdash x = x$.

iv. So, by instance of the tautology

$$(A \Rightarrow (B \Rightarrow C)) \Rightarrow (B \Rightarrow (A \Rightarrow C))$$

and two applications of MP, we get $\vdash x = y \Rightarrow y = x$.

- v. Two applications of Gen yields $\vdash (\forall x)(\forall y)(x = y \Rightarrow y = x)$.
- vi. Then two applications of rule A4 yields $\vdash t = s \Rightarrow s = t$.
- c. i. Let x, y , and z be three variables not occurring in t, s , or r .
ii. Then let $\mathcal{B}(y, y)$ be $y = z$ and $\mathcal{B}(x, x)$ be $x = z$ in (A7).
iii. Now, with x and y interchanged, we obtain $\vdash y = x \Rightarrow (y = z \Rightarrow x = z)$.
iv. By, by (b), $\vdash x = y \Rightarrow y = x$.
v. Now, using an instance of the tautology

$$(A \Rightarrow B) \Rightarrow ((B \Rightarrow C) \Rightarrow (A \Rightarrow C))$$

and two applications of MP, we obtain $\vdash x = y \Rightarrow (y = z \Rightarrow x = z)$.

vi. Similarly to our proof of (b), by three applications of gen we obtain

$$(\forall x)(\forall y)(\forall z)(x = y \Rightarrow (y = z \Rightarrow x = z))$$

vii. Then three applications of rule A4 obtains $\vdash t = s \Rightarrow (s = r \Rightarrow t = r)$.

Proposition 2.24 Let K be a theory for which (A6) holds and (A7) holds for all atomic wfs $\mathcal{B}(x, x)$ in which there are no individual constants. Then K is a theory with equality, that is, (A7) holds for all wfs $\mathcal{B}(x, x)$.

Proof This proposition serves the purpose of reducing the schema (A7) to simpler cases. I don't feel like writing this proof up at the moment and it doesn't seem incredibly significant so I am skipping it for now.

Proposition 2.25 Let K be any theory in which (A6) holds and the following are true:

- Schema (A7) holds for all atomic wfs $\mathcal{B}(x, x)$ such that no function letters or individual constants occur in $\mathcal{B}(x, x)$ and $\mathcal{B}(x, y)$ comes from $\mathcal{B}(x, x)$ by replacing exactly one occurrence of x by y .
- $\vdash x = y \Rightarrow f_j^n(z_1, \dots, z_n) = f_j^n(w_1, \dots, w_n)$, where f_j^n is any function letter of K , $z_1, \dots, z_n$ are variables, and $f_j^n(w_1, \dots, w_n)$ arises from $f_j^n(z_1, \dots, z_n)$ by replacing exactly one occurrence of x by y .

Then K is a theory of equality.

Proof

- By repeated application, our assumptions can be extended to replacements of one or more than one occurrence of x by y .
- Also, Proposition 2.23 is still derivable.
- By Proposition 2.24, it suffices to prove (A7) for only atomic wfs without individual constants.
- But, hypothesis (a) allows us to prove

$$\vdash (y_1 = z_1 \wedge \dots \wedge y_n = z_n) \Rightarrow (\mathcal{B}(y_1, \dots, y_n) \Rightarrow \mathcal{B}(z_1, \dots, z_n))$$

for all variables $y_1, \dots, y_n, z_1, \dots, z_n$ and any atomic wf $\mathcal{B}(y_1, \dots, y_n)$ without function letters or individual constants.

- So, it suffices to show the following:

(*) If $t(x, x)$ is a term without individual constants and $t(x, y)$ comes from $t(x, x)$ by replacing some occurrences of x by y , then $\vdash x = y \Rightarrow t(x, x) = t(x, y)$.⁴

- But (*) can be proved using hypothesis (b) by induction on the number of function letters in $t(x, x)$. This is left as an exercise in the book, maybe I will add this later (probably not, sorry).
- It can be seen from Proposition 2.25 that it is only necessary to verify (A7) for a finite list of special cases when the language K has only finitely many predicate and function letters.⁵

Some Additional notes. Not sure why this isn't a proposition, considering that this part of the textbook seems very much like a proof.

⁴ You can clarify how (*) is applied by using it to prove the following instance of (A7): $\vdash x = y \Rightarrow (A_1^1(f_1^1(x)) \Rightarrow A_1^1(f_1^1(y)))$. Let $t(x, x)$ be $f_1^1(x)$ and let $t(x, y)$ be $f_1^1(y)$.

⁵ In fact, n wfs for each A_j^n and n wfs for each f_j^n .

1. In any model for a theory K with equality, the relation E in the model corresponding to the predicate letter = is an equivalence relation (by Proposition 2.23).
2. If this relation E is the identity relation in the domain of the model, then the model is said to be *normal*.
3. Any model M for K can be *contracted* to a normal model M* for K by taking the domain D* of M* to be the set of equivalence classes determined by the relation E in the domain D of M.
4. For a predicate letter A_j^n and for any equivalence classes $[b_1], \dots, [b_n]$ in D* determined by elements $b_1, \dots, b_n$ in D, we let $(A_j^n)^{M^*}$ hold for $([b_1], \dots, [b_n])$ iff A_j^n holds for $(b_1, \dots, b_n)$.
5. Notice that it makes no difference which representatives $b_1, \dots, b_n$ we select in the given equivalence classes because, from (A7):

$$\vdash x_1 = y_1 \wedge \dots \wedge x_n = y_n \Rightarrow (A_j^n(x_1, \dots, x_n) \Leftrightarrow A_j^n(y_1, \dots, y_n))$$

6. Likewise, for any function letter f_j^n and any equivalence classes $[b_1], \dots, [b_n]$ in D*, let $(f_j^n)^{M^*}([b_1], \dots, [b_n]) = [f_j^n(b_1, \dots, b_n)]$.
7. Again note that this is independent of the choice of the representatives $b_1, \dots, b_n$, since, from (A7), we can prove:

$$\vdash x_1 = y_1 \wedge \dots \wedge x_n = y_n \Rightarrow (f_j^n(x_1, \dots, x_n) \Leftrightarrow f_j^n(y_1, \dots, y_n))$$

8. For any individual constant a_i let $(a_i)^{M^*} = [a_i]$.
9. The relation E* corresponding to = in the model M* is the identity relation in D* : $E^*([b_1], [b_2])$ iff $E(b_1, b_2)$, that is, iff $[b_1] = [b_2]$.
10. Now one can prove by induction the following lemma:
11. If $s = (b_1, b_2, \dots)$ is a denumerable sequence of equivalence classes, then a wf $\mathcal{B}$ is satisfied by s in M iff $\mathcal{B}$ is satisfied by s' in M*.
12. It follows that, for any wf $\mathcal{B}$, $\mathcal{B}$ is true for M iff $\mathcal{B}$ is true for M*.
13. Hence, because M is a model of K, M* is a normal model of K.

Proposition 2.26 (Extension of Proposition 2.17) (Gödel, 1930) Any consistent theory with equality K has a finite or denumerable normal model.

Proof

By Proposition 2.17, K has a denumerable model M. Hence, the contraction of M to a normal model yields a finite or denumerable normal model M* because the set of equivalence classes in a denumerable set D is either finite or denumerable.

Corollary 2.27 (Extension of Skolem-Löwenheim Theorem) Any theory with equality K that has an infinite normal model M has a denumerable normal model.

Proof

1. Add to K the denumerably many new individual constants $b_1, b_2, \dots$ together with the axioms $b_i \neq b_j$ for $i \neq j$

2. Then the new theory K' is consistent.
3. If K' were inconsistent, there would be a proof in K' of a contradiction $\mathcal{C} \wedge \neg\mathcal{C}$, where we may assume that $\mathcal{C}$ is a wf of K .⁶
4. But this proof uses only a finite number of the new axioms: $b_{i1} \neq b_{j1}, \dots, b_{in} \neq b_{jn}$.
5. Now, M can be extended to a model $M^\#$ of K plus the axioms $b_{i1} \neq b_{j1}, \dots, b_{in} \neq b_{jn}$.
6. In fact, since M is an infinite normal model, we can choose interpretations of $b_{i1}, b_{j1}, \dots, b_{in}, b_{jn}$ so that the wfs $b_{i1} \neq b_{j1}, \dots, b_{in} \neq b_{jn}$ are true.
7. But, since $\mathcal{C} \wedge \neg\mathcal{C}$ is derivable from these wfs and the axioms of K , it would follow that $\mathcal{C} \wedge \neg\mathcal{C}$ is true for $M^\#$, which is impossible.
8. Hence, K' must be consistent.
9. Now, by Proposition 2.26, K' has a finite or denumerable normal model N .
10. But, since, for $i \neq j$, the wfs $b_i \neq b_j$ are axioms of K' , they are true for N .
11. Thus, the elements in the domain of N that are interpretations of $b_1, b_2, \dots$ must be distinct, which implies that the domain of N is infinite, and therefore, denumerable.

⁶ If K' is inconsistent, it can prove anything, including a wf $\mathcal{C}$ of K and its negation $\neg\mathcal{C}$.

3 Formal Number Theory

3.1 An Axiom System

We build our first-order theory S using our language of arithmetic, $\mathcal{L}_A$. It has a single predicate letter A_1^2 . Because we want a normal interpretation, we write $t = s$ for $A_1^2(t, s)$. $\mathcal{L}_A$ has one individual constant a_1 . 0 is the shorthand for a_1 . Finally, $\mathcal{L}_A$ has three function letters, f_1^1 , f_1^2 , and f_2^2 .

The proper axioms of S are:

$$(S1) \quad x_1 = x_2 \Rightarrow (x_1 = x_3 \Rightarrow x_2 = x_3)$$

$$(S2) \quad x_1 = x_2 \Rightarrow x'_1 = x'_2$$

$$(S3) \quad 0 \neq x'_1$$

$$(S4) \quad x'_1 = x'_2 \Rightarrow x_1 = x_2$$

$$(S5) \quad x_1 + 0 = x_1$$

$$(S6) \quad x_1 + x_2 = (x_1 + x_2)'$$

$$(S7) \quad x_1 \cdot 0 = 0$$

$$(S8) \quad x_1 \cdot (x_2)' = (x_1 \cdot x_2) + x_1$$

$$(S9) \quad \mathcal{B}(0) \Rightarrow ((\forall x)(\mathcal{B}(x) \Rightarrow \mathcal{B}(x')) \Rightarrow (\forall x)\mathcal{B}(x)) \text{ for any wf } \mathcal{B}(x) \text{ of S.}$$

(S9) represents the *principle of mathematical induction*. Notice that (S1)-(S8) are particular wfs, while (S9) is an axiom schema.

From (S9) by MP, we obtain the *induction rule*:

$$\mathcal{B}(0), (\forall x)(\mathcal{B}(x) \Rightarrow \mathcal{B}(x')) \vdash_s (\forall x)\mathcal{B}(x)$$

Lemma 3.1

For any terms t, s, r , of $\mathcal{L}_A$, the following wfs are theorems of S.

$$(S1)' \quad t = r \Rightarrow (t = s \Rightarrow r = s)$$

$$(S2)' \quad t = r \Rightarrow t' = r'$$

$$(S3)' \quad 0 \neq t'$$

$$(S4)' \quad t' = r' \Rightarrow t = r$$

$$(S5)' \quad t + 0 = t$$

$$(S6)' \quad t + r = (t + r)'$$

$$(S7)' \quad t \cdot 0 = 0$$

$$(S8)' \quad t \cdot (r)' = (t \cdot r) + t$$

Proof

(S1)'-(S8)' follow from (S1)-(S8), respectively. First for the closure by means of Gen, use Exercise 2.48⁷ to change all the bound variables to variables not occurring in terms t, r, s , and then apply rule A4 with the appropriate terms t, r, s .

Proposition 3.2

For any terms t, s, r , the following wfs are theorems of S.

- a. $t = t$
- b. $t = r \Rightarrow r = t$
- c. $t = r \Rightarrow (r = s \Rightarrow t = s)$
- d. $r = t \Rightarrow (s = t \Rightarrow r = s)$
- e. $t = r \Rightarrow t + s = r + s$
- f. $t = 0 + t$
- g. $t' + r = (t + r)'$
- h. $t + r = r + t$
- i. $t = r \Rightarrow s + t = s + r$
- j. $(t + r) + s = t + (r + s)$
- k. $t = r \Rightarrow t \cdot s = r \cdot s$
- l. $0 \cdot t = 0$
- m. $t' \cdot r = t \cdot r + r$
- n. $t \cdot r = r \cdot t$
- o. $t = r \Rightarrow s \cdot t = s \cdot r$

Proof (The rest are in the textbook)

- a.
 - 1. $t + 0 = t$ (S5)'
 - 2. $(t + 0 = t) \Rightarrow (t + 0 = t \Rightarrow t = t)$ (S1)'
 - 3. $t + 0 = t \Rightarrow t = t$ 1, 2, MP
 - 4. $t = t$ 1, 3, MP
- g. Let $\mathcal{B}(y)$ be $x' + y = (x + y)'$.
 - i. 1. $x' + 0 = x'$ (S5')
 - 2. $x + 0 = x$ (S5')
 - 3. $(x + 0)' = x'$ 2, (S2'), MP

⁷ Exercise 2.48: If $\mathcal{B}(x)$ is similar to $\mathcal{B}(y)$, $(\forall x)\mathcal{B}(x)$ is a subformula of $\mathcal{C}$, and $\mathcal{C}'$ is the result of replacing one or more occurrences of $(\forall x)\mathcal{B}(x)$ in $\mathcal{C}$ by $(\forall y)\mathcal{B}(y)$, prove that $\vdash \mathcal{C} \Leftrightarrow \mathcal{C}'$.

$$4. \quad x' + 0 = (x + 0)' \quad 1, 3, \text{ part (d), MP}$$

Thus, $\vdash_S \mathcal{B}(0)$

- ii. 1. $x' + y = (x + y)'$ Hyp
 2. $x' + y' = (x' + y)'$ (S6')
 3. $(x' + y')' = (x + y)''$ 1, (S2'), MP
 4. $x' + y' = (x + y)''$ 2, 3, part (c), MP
 5. $x + y' = (x + y)'$ (S6')
 6. $(x + y')' = (x + y)''$ 5, (S2'), MP
 7. $x' + y' = (x + y)'$ 4, 6, part (d), MP
 8. $\vdash_S x' + y = (x + y)' \Rightarrow x' + y' = (x + y)'$ 1-7, deduction theorem

Thus, $\vdash_S \mathcal{B}(y) \Rightarrow \mathcal{B}(y')$, and, by Gen, $\vdash_S (\forall y)(\mathcal{B}(y) \Rightarrow \mathcal{B}(y'))$. Hence, by (i), (ii), and the induction rule, $\vdash_S (\forall y)(x' + y = (x + y)')$. By Gen and rule A4, $\vdash_S t' + r = (t + r)'$

Corollary 3.3

S is a theory of with equality

Proof

By Prop 2.25, this reduces to parts (a)-(e), (i), (k), and (o) of Prop 3.2, and (S2').

Notice that the interpretation in which

- The set of nonnegative integers is the domain
- The integer 0 is the interpretation of the symbol 0
- The successor operation (addition of 1) is the interpretation of the function (that is, of f_1^1)
- Ordinary addition and multiplication are the interpretations of $+$ and $\cdot$
- The interpretation of the predicate letter $=$ is the identity relation

is a normal model for S. This is the *standard interpretation* or *standard model*. Any normal model for S which is not isomorphic to the standard model will be called a *nonstandard* model of S.

It is important to note that if we recognize the standard interpretation to be a model for S, then S is consistent. However, this is somewhat of a semantic argument and is regarded by some as too precarious to serve as a basis for consistency proofs. Mendelson doesn't prove that the axioms of S are true under the standard interpretation, but it is taken to be intuitively obvious. It is common to take the statement of the consistency of S as an explicit unproven assumption.

Proposition 3.4

For and terms t, r, s , the following wfs are theorems of S.

- $t \cdot (r + s) = (t \cdot r) + (t \cdot s)$ (distributivity)
- $(r + s) \cdot t = (r \cdot t) + (s \cdot t)$ (distributivity)

- c. $(t \cdot r) \cdot s = t \cdot (r \cdot s)$ (associativity of $\cdot$)
- d. $t + s = r + s \Rightarrow t = r$ (cancellation law for $+$)

Proof

- a. Prove $\vdash_S x \cdot (y + z) = (x \cdot y) + (x \cdot z)$ by induction on z .
- b. Use part (a) and Proposition 3.2(n).
- c. Prove $\vdash_S (x \cdot y) \cdot z = x \cdot (y \cdot z)$ by induction on z .
- d. Prove $\vdash_S x + z = y + z \Rightarrow x = y$ by induction on z . This requires, for the first time, use of (S4').

The terms $0, 0', 0'', 0''', \dots$ we shall call *numerals* and denote by $\bar{0}, \bar{1}, \bar{2}, \bar{3}, \dots$. More precisely, $\bar{0}$ is 0 and, for any natural number n , $\overline{n+1}$ is $(\bar{n})'$. Essentially, $\bar{n}$ is the shorthand for a natural number n which stands for a 0 with n successors applied. That is:

$$\underbrace{0''''\dots}_{n \text{ times}} = \bar{n} \quad \text{which represents the numeral } n \text{ in the metalanguage.}$$

Proposition 3.5

The following are theorems of S.

- a. $t + \bar{1} = t'$
- b. $t \cdot \bar{1} = t$
- c. $t \cdot \bar{2} = t + t$
- d. $t + s = 0 \Rightarrow t = 0 \wedge s = 0$
- e. $t \neq 0 \Rightarrow (s \cdot t = 0 \Rightarrow s = 0)$
- f. $t + s = \bar{1} \Rightarrow (t = 0 \wedge s = \bar{1}) \vee (t = \bar{1} \wedge s = 0)$
- g. $t \cdot s = \bar{1} \Rightarrow (t = \bar{1} \wedge s = \bar{1})$
- h. $t \neq 0 \Rightarrow (\exists y)(t = y')$
- i. $s \neq 0 \Rightarrow (t \cdot s = r \cdot s \Rightarrow t = r)$
- j. $t \neq 0 \Rightarrow (t \neq 1 \Rightarrow (\exists y)(t = y''))$

Proof

The proofs in the chapter are pretty clear for this one so I'll spare myself the effort of typing it up.

Proposition 3.6

- a. Let m and n be any natural numbers.
 - i. If $m \neq n$, then $\vdash_S \bar{m} \neq \bar{n}$.
 - ii. $\vdash_S \overline{m+n} = \bar{m} + \bar{n}$ and $\overline{m \cdot n} = \bar{m} \cdot \bar{n}$.
- b. Any model for S is infinite.
- c. For any cardinal number $\aleph_\beta$, S has a normal model of cardinality $\aleph_\beta$.

Proof

- a. i. Assume $m \neq n$. Either $m < n$ or $n < m$. Suppose, without loss of generality, that $m < n$.⁸

1. $\overline{m} = \overline{n}$ Hyp
2. $0 \overbrace{''' \dots}^{m \text{ times}} = 0 \overbrace{''' \dots}^{n \text{ times}}$ 1 is an abbreviation of 2
3. Apply (S4') and MP m times. We get $0 = 0 \overbrace{''' \dots}^{n-m \text{ times}}$.
Let $t = n - m - 1$. Since $n > m$, $n - m - 1 \geq 0$.
Thus, we obtain $0 = t'$.
4. $0 \neq t'$ (S3')
5. $0 = t' \wedge 0 \neq t'$ 3, 4, $\wedge$ intro
6. $\overline{m} = \overline{n} \vdash_S 0 = t' \wedge 0 \neq t'$ 1-5
7. $\vdash_S \overline{m} \neq \overline{n}$ 1-6, proof by $\perp$

- ii. We use induction in the metalanguage.

1. First, $\overline{m} + \overline{0}$ is $\overline{m}$.
2. Hence, $\vdash_S \overline{m} + \overline{0} = \overline{m} + \overline{0}$ by (S5').
3. Now assume $\vdash_S \overline{m} + \overline{n} = \overline{m} + \overline{n}$.
4. Then $\vdash_S (\overline{m} + \overline{n})' = \overline{m} + (\overline{n})'$ by (S2') and (S6').
5. But, $\overline{m} + (\overline{n} + \overline{1})$ is $(\overline{m} + \overline{n})'$ and $\overline{n} + \overline{1}$ is $(\overline{n})'$.
6. Hence, $\vdash_S \overline{m} + (\overline{n} + \overline{1}) = \overline{m} + (\overline{n} + \overline{1})$.
7. Thus, $\vdash_S \overline{m} + \overline{n} = \overline{m} + \overline{n}$.
8. The proof that $\vdash_S \overline{m} \cdot \overline{n} = \overline{m} \cdot \overline{n}$ is left as an exercise for the reader.⁹

- b. By part (a), (i), in a model for S the objects corresponding to the numerals must be distinct. But there are denumerably many numerals.
- c. This follows from Corollary 2.34(c) and the fact that the standard model is an infinite normal model.

Now we can introduce an order relation in S:

$$\begin{aligned}
 t < s & \text{ for } (\exists w)(w \neq 0 \wedge w + t = s) \\
 t \leq s & \text{ for } t < s \vee t = s \\
 t > s & \text{ for } s < t \\
 t \geq s & \text{ for } s \leq t \\
 t \not< s & \text{ for } \neg(t < s), \text{ and so on}
 \end{aligned}$$

⁸ That is, if it is actually the case that $n < m$ then switch them around.

⁹ For those taking (or who have taken) CS 374, notice how similar this is to string induction in the first few weeks!

In the first definition, we choose w to be the first variable not in t or s .

Proposition 3.7

For any terms t, r, s , the following are theorems. Refer to page 163 of the textbook for these, nothing exciting or unclear here (e.g. $t \not\prec t, r \neq 0 \Leftrightarrow r > 0$, etc.).

Proposition 3.8

- a. For any natural number k , $\vdash_S x = 0 \vee \dots \vee x = \bar{k} \Leftrightarrow x \leq \bar{k}$.
- a'. For any natural number k and any wf $\mathcal{B}$, $\vdash_S \mathcal{B}(0) \wedge \mathcal{B}(\bar{1}) \wedge \dots \wedge \mathcal{B}(\bar{k}) \Leftrightarrow (\forall x)(x \leq \bar{k} \Rightarrow \mathcal{B}(x))$.
- b. For any natural number $k > 0$, $\vdash_S x = 0 \vee \dots \vee x = \overline{(k-1)} \Leftrightarrow x < \bar{k}$.
- b'. For any natural number $k > 0$ and any wf $\mathcal{B}$, $\vdash_S \mathcal{B}(0) \wedge \mathcal{B}(\bar{1}) \wedge \dots \wedge \mathcal{B}(\overline{(k-1)}) \Leftrightarrow (\forall x)(x < \bar{k} \Rightarrow \mathcal{B}(x))$.
- c. $\vdash_S ((\forall x)(x < y \Rightarrow \mathcal{B}(x)) \wedge (\forall x)(x \geq y \Rightarrow \mathcal{C}(x))) \Rightarrow (\forall x)(\mathcal{B}(x) \vee \mathcal{C}(x))$.

Proof

- a. We prove $\vdash_S x = 0 \wedge \dots \wedge x = \bar{k} \Leftrightarrow x \leq \bar{k}$ by induction in the metalanguage on k .
 - 1. The base case of $k = 0$, $\vdash_S x = 0 \Leftrightarrow x \leq$ is easy to get from the definitions and Proposition 3.7.
 - 2. **IH:** Assume that $\vdash_S x = 0 \vee \dots \vee x = \bar{k} \Leftrightarrow x \leq \bar{k}$.
 - 3. Now, assume $x = 0 \vee \dots \vee x = \bar{k} \vee x = \overline{(k+1)}$.
 - 4. But, $\vdash_S x = \overline{(k+1)} \Rightarrow x \leq \overline{(k+1)}$ and, by **IH**, $\vdash_S x = 0 \vee \dots \vee x = \bar{k} \Leftrightarrow x \leq \bar{k}$.
 - 5. Also, $\vdash_S x \leq \bar{k} \Rightarrow x \leq \overline{(k+1)}$.
 - 6. Thus, $x \leq \overline{(k+1)}$.
 - 7. So, $\vdash_S x = 0 \vee \dots \vee x = \bar{k} \vee x = \overline{(k+1)} \Leftrightarrow x \leq \overline{(k+1)}$.
 - 8. Conversely, assume that $x \leq \overline{(k+1)}$.
 - 9. Then, $x = \overline{(k+1)} \vee x < \overline{(k+1)}$.
 - 10. If $x = \overline{(k+1)}$, then $x = 0 \vee \dots \vee x = \bar{k} \vee x = \overline{(k+1)}$.
 - 11. If $x < \overline{(k+1)}$, then because $\overline{(k+1)}$ is $(\bar{k})'$, we have $x \leq \bar{k}$ by Proposition 3.7(1).
 - 12. By **IH**, $x = 0, \vee \dots \vee x = \bar{k}$, and therefore $x = 0 \vee \dots \vee x = \bar{k} \vee x = \overline{(k+1)}$.
 - 13. In either case, $x = 0 \vee \dots \vee x = \bar{k} \vee x = \overline{(k+1)}$.
 - 14. This proves $\vdash_S x \leq \overline{(k+1)} \Rightarrow x = 0 \vee \dots \vee x = \bar{k} \vee x = \overline{(k+1)}$.
 - 15. From **IH**, we have derived that $\vdash_S x = 0 \vee \dots \vee x = \overline{(k+1)} \Leftrightarrow x \leq \overline{(k+1)}$ and this completes the proof.

Here and onwards Mendelson begins to give more informal proofs as you may have noticed. He begins to omit deduction theorem, rule C, replacement theorem, tautologies, etc. Also, he does not provide proofs for the other theorems.

Proposition 3.9 There are other forms of the induction principle which can now be formulated.

- a. *Complete Induction*: $\vdash_S (\forall x)((\forall z)(z < x \Rightarrow \mathcal{B}(z)) \Rightarrow \mathcal{B}(x)) \Rightarrow (\forall x)\mathcal{B}(x)$. To translate to ordinary language, consider a property P such that, for any x , if P holds for all natural numbers less than x , then P holds for x also. Then P holds for all natural numbers.
- b. *Least-number Principle*: $\vdash_S (\exists x)\mathcal{B}(x) \Rightarrow (\exists y)\mathcal{B}(y) \wedge (\forall z)(z < y \Rightarrow \neg \mathcal{B}(z))$. If a property P holds for some natural number, then there is a least number satisfying P .

a. Let $\mathcal{C}(x)$ be $(\forall z)(z \leq x \Rightarrow \mathcal{B}(z))$.

- | | | | |
|-----|----|---|---|
| i. | 1. | $(\forall x)((\forall z)(z < x \Rightarrow \mathcal{B}(z)) \Rightarrow \mathcal{B}(x))$ | Hyp |
| | 2. | $(\forall z)(z < 0 \Rightarrow \mathcal{B}(z)) \Rightarrow \mathcal{B}(0)$ | 1, rule A4 |
| | 3. | $z \not< 0$ | Proposition 3.7(y) |
| | 4. | $(\forall z)(z < 0 \Rightarrow \mathcal{B}(z))$ | 3, tautology, Gen |
| | 5. | $\mathcal{B}(0)$ | 2, 4 MP |
| | 6. | $(\forall z)(z \leq 0 \Rightarrow \mathcal{B}(z))$, i.e., $\mathcal{C}(0)$ | 5, Proposition 3.8(a') |
| | 7. | $(\forall x)((\forall z)(z < x \Rightarrow \mathcal{B}(z)) \Rightarrow \mathcal{B}(x)) \vdash_S \mathcal{C}(0)$ | 1-6 |
| ii. | 1. | $(\forall x)((\forall z)(z < x \Rightarrow \mathcal{B}(z)) \Rightarrow \mathcal{B}(x))$ | Hyp |
| | 2. | $\mathcal{C}(x)$, i.e., $(\forall z)(z \leq x \Rightarrow \mathcal{B}(z))$ | Hyp |
| | 3. | $(\forall z)(z < x' \Rightarrow \mathcal{B}(z))$ | 2, Proposition 3.7(ℓ) |
| | 4. | $(\forall z)(z < x' \Rightarrow \mathcal{B}(z)) \Rightarrow \mathcal{B}(x')$ | 1, rule A4 |
| | 5. | $\mathcal{B}(x')$ | 3, 4, MP |
| | 6. | $z \leq x \Rightarrow z < x' \vee z = x'$ | Definition, tautology |
| | 7. | $z < x' \Rightarrow \mathcal{B}(z)$ | 3, rule A4 |
| | 8. | $z = x' \Rightarrow \mathcal{B}(z)$ | 5, (A7), Proposition 2.23(b), tautologies |
| | 9. | $(\forall z)(z \leq x' \Rightarrow \mathcal{B}(z))$, i.e., $\mathcal{C}(x')$ | 6, 7, 8, Tautology, Gen |
| 10. | | $(\forall x)((\forall z)(z < x \Rightarrow \mathcal{B}(z)) \Rightarrow \mathcal{B}(x))$
$\vdash_S (\forall x)(\mathcal{C}(x) \Rightarrow \mathcal{C}(x'))$ | 1-9, ded. them., Gen |

By (i), (ii), and the induction rule, we obtain $\mathcal{D} \vdash_S (\forall x)(\mathcal{C}(x))$, that is, $\mathcal{D} \vdash_S (\forall x)(\forall z)(z \leq x \Rightarrow \mathcal{B}(z))$, where $\mathcal{D}$ is $(\forall x)((\forall z)(z < x \Rightarrow \mathcal{B}(z)) \Rightarrow \mathcal{B}(x))$. Hence, by rule A4 twice, $\mathcal{D} \vdash_S x \leq x \Rightarrow \mathcal{B}(x)$. Notice that $\vdash_S x \leq x$. So, $\mathcal{D} \vdash_S \mathcal{B}(x)$, and, by Gen and deduction theorem, $\vdash_S \mathcal{D} \Rightarrow (\forall x)\mathcal{B}(x)$.

b. Skipping this one.

Definition

$t \mid s$ for $(\exists z)(s = t \cdot z)$. (Here, z is the first variable now in t or s).

Proposition 3.10 The following wfs are theorems for any terms t , s , and r .

- a. $t \mid t$
- b. $1 \mid t$
- c. $t \mid 0$
- d. $t \mid s \wedge s \mid r \Rightarrow t \mid r$
- e. $s \neq 0 \wedge t \mid s \Rightarrow t \leq s$
- f. $t \mid s \wedge s \mid t \Rightarrow s = t$
- g. $t \mid s \Rightarrow t \mid (r \cdot s)$
- h. $t \mid s \wedge t \mid r \Rightarrow t \mid (s + r)$

Proof Skipping, the proofs aren't quite as important for these notions of math at this point. They are purely for rigor in confirming that we are able to prove these arithmetical definitions to allow us to set things up for later proofs, e.g. Gödel numbering.

Proposition 3.11

$$\vdash_S y \neq 0 \Rightarrow (\exists u)(\exists v)[x = y \cdot u + v \wedge c < y \wedge (\forall u_1)(\forall v_1)((x = y \cdot u_1 + v_1 \wedge v_1 < y) \Rightarrow y = u_1 \wedge v = v_1)]$$

Proof «Weird formatting error here for step 1 in the textbook»

- i.
 - 1. $y \neq 0$ Hyp
 - 2. $0 = y \cdot 0 + 0$ (S5'), (S7')
 - 3. $0 < y$ 1, Proposition 3.7(t)
 - 4. $0 = y \cdot 0 + 0 \wedge 0 < y$ 2, 3, conjunction rule
 - 5. $(\exists u)(\exists v)(0 = y \cdot u + v \wedge v < y)$ 4, rule E4 twice
 - 6. $y \neq 0 \Rightarrow (\exists u)(\exists v)(0 = y \cdot u + v \wedge v < y)$ 1-5, ded. them.
- ii.
 - 1. $\mathcal{B}(x)$, i.e., $y \neq 0 \Rightarrow (\exists u)(\exists v)(x = y \cdot u + v \wedge v < y)$ Hyp
 - 2. $y \neq 0$ Hyp
 - 3. $(\exists u)(\exists v)(x = y \cdot u + v \wedge v < y)$ 1, 2, MP
 - 4. $x = y \cdot a + b \wedge b < y$ 4, rule C twice
 - 5. $b < y$ 4, $\wedge$ elim
 - 6. $b' \leq y$ 5, Proposition 3.7(k)
 - 7. $b' < y \vee b' = y$ 6, definition
 - 8. $b' < y \Rightarrow (x' = y \cdot a + b' \wedge b' < y)$ 4, (S6'), derived rules
 - 9. $b' < y \Rightarrow (\exists u)(\exists v)(x' = y \cdot u + v \wedge v < y)$ 8, rule E4, ded. them.
 - 10. $b' = y \Rightarrow x' = y \cdot a + y \cdot \bar{1}$ 4, (S6'), Proposition 3.5(b)
 - 11. $b' = y \Rightarrow (x' = y \cdot (a + \bar{1}) + 0 \wedge 0 < y)$ 10, Proposition 3.7(t), (S5')
 - 12. $b' = y \Rightarrow (\exists u)(\exists v)(x' = y \cdot u + v \wedge v < y)$ 11, rule E4 twice, ded. them.

13. $(\exists u)(\exists v)(x' = y \cdot u + v \wedge v < y)$ 7, 9, 12, $\vee$ elim
14. $\mathcal{B}(x) \Rightarrow (y \neq 0 \Rightarrow (\exists u)(\exists v)(x' = y \cdot u + v \vee v < y))$, i.e., $\mathcal{B}(x) \Rightarrow \mathcal{B}(x')$ 1-13, ded. them.

By (i), (ii), Gen, and the induction rule, $\vdash_S (\forall x)(\mathcal{B}(x))$. This establishes the existence of a quotient u and a remainder v . To prove uniqueness, proceed as follows:

1. Assume $y \neq 0$.
2. Assume $x = y \cdot u + v \wedge v < y$ and $x = y \cdot u_1 + v_1 \wedge v_1 < y$.
3. **«might be a mistake above, noticed latex issue but unsure»**
4. Now, $u = u_1$ or $u < u_1$ or $u_1 < u$.
5. If $u = u_1$, then $v = v_1$ by Proposition 3.4(d).
6. If $u < u_1$, then $u_1 = u_w$ for some $w \neq 0$.
7. Then $y \cdot u + v = y \cdot (u + w) + v_1 = y \cdot u + y \cdot w + v_1$.
8. Hence, $v = y \cdot w + v_1$.
9. Since $w \neq 0$, $y \cdot w \geq y$.
10. So, $v = y \cdot w + v_1 \geq y$, contradicting $v < y$.
11. Hence, $u \not< u_1$.
12. Similarly, $u_1 \not< u$.
13. Thus, $u = u_1$.
14. Since $y \cdot u + v = x = y \cdot u_1 + v_1$, it follows that $v = v_1$.

From this point on, one can generally translate into S and prove the results from any text on elementary number theory. There are a few important number-theoretic functions left to define, e.g. x^y and $x!$, which will be done soon. It is important to note that there are some standard results of number theory which are proved with the aid of the theory of complex variables, which we will not provide in S. There are also some statements which rely on non-elementary concepts (e.g. logarithms) which, except in special cases, cannot be expressed in S. However, our system S will still be strong enough to derive our incompleteness results later on in this chapter. Moreover, we will see that the incompleteness of S cannot be attributed to the omission of some essential axiom of, but rather that it has deeper underlying causes that apply to other similarly strong theories as well.

3.2 Number-Theoretic Functions and Relations

Number-theoretic function: function whose arguments and values are natural numbers, e.g. addition and multiplication.

Number-theoretic relation: relation whose arguments are natural numbers, e.g. $=$ and $<$, with $x + y < z$ being a three argument number-theoretic relation.

Let K be any theory in the language $\mathcal{L}_A$ of arithmetic. A number-theoretic relation R of n arguments is *expressible* in K iff there is a wf $\mathcal{B}(x_1, \dots, x_n)$ of K with the free variables $x_1, \dots, x_n$ such that, for any natural numbers $k_1, \dots, k_n$, the following hold:

1. If $R(k_1, \dots, k_n)$ is true, then $\vdash_K \mathcal{B}(\bar{k}_1, \dots, \bar{k}_n)$.
2. If $R(k_1, \dots, k_n)$ is false, then $\vdash_K \neg \mathcal{B}(\bar{k}_1, \dots, \bar{k}_n)$.

For example, the number-theoretic relation of identity is expressed in S by the wf $x_1 = x_2$. Moreover, if $k_1 = k_2$, then $\bar{k}_1$ is the same term as $\bar{k}_2$ and so $\vdash_S \bar{k}_1 = \bar{k}_2$ by Proposition 3.2(a). Similarly, the relation "less than" is expressible in S by the wf $x_1 < x_2$. The proof for this is outlined on page 170.

If a relation is expressible in a theory K , then it is expressible in any extension of K .

Let K be any theory with equality in the language $\mathcal{L}_A$ of arithmetic. A number-theoretic function f of n arguments is said to be *representable* in K iff there is a wf $\mathcal{B}(x_1, \dots, x_n, y)$ of K with the free variables $x_1, \dots, x_n, y$ such that, for any natural numbers $k_1, \dots, k_n, m$, the following hold:

1. If $f(k_1, \dots, k_n) = m$, then $\vdash_K \mathcal{B}(\bar{k}_1, \dots, \bar{k}_n, \bar{m})$.
2. $\vdash_K (\exists_1 y) \mathcal{B}(\bar{k}_1, \dots, \bar{k}_n, y)$

Furthermore, a function f is said to be *strongly representable* in K if we replace condition 2 with the wf $\vdash_K (\exists_1 y) \mathcal{B}(x_1, \dots, x_n, y)$. Notice that this wf implies the second condition by Gen and rule A4. Hence, strong representability implies representability. The converse is also true.

Proposition 3.12 (V.H. Dyson)

If $f(x_1, \dots, x_n)$ is representable in K , then it is strongly representable in K .

Proof

Assume f is representable in K by a wf $\mathcal{B}(x_1, \dots, x_n, y)$. Let us show that f is strongly representable in K by the following wf $\mathcal{C}(x_1, \dots, x_n, y)$:

$$((\exists_1 y) \mathcal{B}(x_1, \dots, x_n, y)) \vee \mathcal{B}(x_1, \dots, x_n, y) \vee (\neg[(\exists_1 y) \mathcal{B}(x_1, \dots, x_n, y)] \vee y = 0)$$

1. Assume $f(k_1, \dots, k_n) = m$. Then $\vdash_K \mathcal{B}(\bar{k}_1, \dots, \bar{k}_n, \bar{m})$ and $\vdash_K (\exists_1 y) \mathcal{B}(\bar{k}_1, \dots, \bar{k}_n, y)$. So, by $\wedge$ introduction and $\vee$ introduction, we get $\vdash_K \mathcal{C}(\bar{k}_1, \dots, \bar{k}_n, \bar{m})$.
- 2'. We must show $\vdash_K (\exists_1 y) \mathcal{C}(x_1, \dots, x_n, y)$.

Case 1. Take $(\exists_1 y) \mathcal{B}(x_1, \dots, x_n, y)$ as hypothesis.

- (i) We obtain $\mathcal{B}(x_1, \dots, x_n, b)$ from our hypothesis with Rule C, where b is a new individual constant. Together with our hypothesis and $\wedge$ and $\vee$ introduction, this yields $\mathcal{C}(x_1, \dots, x_n, b)$ and then, by rule E4, $(\exists y) \mathcal{C}(x_1, \dots, x_n, y)$.
- (ii) Assume $\mathcal{C}(x_1, \dots, x_n, u) \vee \mathcal{C}(x_1, \dots, x_n, v)$. From $\mathcal{C}(x_1, \dots, x_n, u)$ and our hypothesis, we obtain $\mathcal{B}(x_1, \dots, x_n, u)$, and, from $\mathcal{C}(x_1, \dots, x_n, v)$ and our hypothesis, we obtain $\mathcal{B}(x_1, \dots, x_n, v)$. Now, from $(\mathcal{B}(x_1, \dots, x_n, u) \wedge \mathcal{B}(x_1, \dots, x_n, v))$ and our hypothesis, we get $u = v$. By the deduction theorem, $(\mathcal{B}(x_1, \dots, x_n, u) \wedge \mathcal{B}(x_1, \dots, x_n, v)) \Rightarrow u = v$.

From (i) and (ii), $(\exists_1 y)\mathcal{C}(x_1, \dots, x_n, y)$. Thus, we have proved $\vdash_K (\exists_1 y)\mathcal{B}(x_1, \dots, x_n, y) \Rightarrow (\exists_1 y)\mathcal{C}(x_1, \dots, x_n, y)$.

Case 2. Take $\neg(\exists_1 y)\mathcal{B}(x_1, \dots, x_n, y)$ as hypothesis.

- (i) From our hypothesis and the theorem $0 = 0$, we obtain $\neg(\exists_1 y)\mathcal{B}(x_1, \dots, x_n, y) \wedge 0 = 0$ by $\wedge$ introduction. By $\vee$ introduction, $\mathcal{C}(x_1, \dots, x_n, 0)$, and, by rule E4, $(\exists y)\mathcal{C}(x_1, \dots, x_n, y)$.
- (ii) Assume $\mathcal{C}(x_1, \dots, x_n, u) \wedge \mathcal{C}(x_1, \dots, x_n, v)$. From $\mathcal{C}(x_1, \dots, x_n, u)$ and our hypothesis, it follows that $u = 0$. Likewise, from $\mathcal{C}(x_1, \dots, x_n, v)$ and our hypothesis, $v = 0$. Hence, $u = v$. By the deduction theorem, we obtain $\mathcal{C}(x_1, \dots, x_n, u) \wedge \mathcal{C}(x_1, \dots, x_n, v) \Rightarrow u = v$.

From (i) and (ii), $(\exists_1 y)\mathcal{C}(x_1, \dots, x_n, y)$. Thus, we have proved $\vdash_K (\exists_1 y)\mathcal{C}(x_1, \dots, x_n, y)$.

Since we have shown them to be equivalent, from now on we shall use representability and strong representability interchangeably.

Observe that a function representable in K is representable in any extension of K .

Examples

In these examples, let K be any theory with equality in the language $\mathcal{L}_A$.

- 1. The zero function, $Z(x) = 0$, is representable by the wf $x_1 = x_1 \wedge y = 0$. For any k and m , if $Z(k) = m$, then $m = 0$ and $\vdash_K \bar{k} = \bar{k} \wedge 0 = 0$ which means that condition 1 holds. Then, we can show that $\vdash_K (\exists_1 y)(x_1 = x_1 \wedge y = 0)$. Thus, condition 2' holds.
- 2. The successor function, $N(x) = x + 1$, is representable in K by the wf $y = x'_1$. For any k and m , if $N(k) = m$, then $m = k + 1$ which means that $\bar{m}$ is $\bar{k}'$. Then $\vdash_K \bar{m} = \bar{k}'$. One can verify that $\vdash_K (\exists_1 y)(y = x'_1)$.
- 3. The projection function, $U_j^n(x_1, \dots, x_n) = x_j$, is representable in K by $x_1 = x_1 \wedge x_2 = x_2 \wedge \dots \wedge x_n = x_n \wedge y = x_j$. If $U_j^n(k_1, \dots, k_n) = m$, then $m = k_j$. Hence, $\vdash_K \bar{k}_1 = \bar{k}_1 \wedge \bar{k}_2$

3.3 Primitive Recursive and Recursive Functions

Definition:

- 1. The following functions are called *initial functions*.
 - I. The *zero function*, $Z(x) = 0$ for all x
 - II. The *successor function*, $N(x) = x + 1$ for all x
 - III. The *projection functions*, $U_i^n(x_1, \dots, x_n) = x_i$ for all $x_1, \dots, x_n$
- 2. The following are rules for obtaining new functions from given functions.
 - IV. *Substitution*:

$$f(x_1, \dots, x_n) = g(h_1(x_1, \dots, x_n), \dots, h_m(x_1, \dots, x_n))$$

f is said to be obtained by substitution from the functions

$$g(y_1, \dots, y_m), h_1(x_1, \dots, x_n), \dots, h_m(x_1, \dots, x_n)$$

V. Recursion:

$$f(x_1, \dots, x_n, 0) = g(x_1, \dots, x_n)$$

$$f(x_1, \dots, x_n, y + 1) = h(x_1, \dots, x_n, y, f(x_1, \dots, x_n, y))$$

Here, we allow $n = 0$, in which case we have:

$$f(0) = k \quad \text{where } k \text{ is a fixed natural number}$$

$$f(y + 1) = h(y, f(y))$$

Proposition 3.16

The following functions are primitive recursive.

a. $x + y$

b. $x \cdot y$

c. x^y

d. $\delta(x) = \begin{cases} x - 1 & \text{if } x > 0 \\ 0 & \text{if } x = 0 \end{cases}$

This is the *predecessor* function

e. $x \dot{-} y = \begin{cases} x - y & \text{if } x \geq y \\ 0 & \text{if } x < y \end{cases}$

f. $\text{sg}(x) = \begin{cases} 0 & \text{if } x = 0 \\ 1 & \text{if } x \neq 0 \end{cases}$

g. $\overline{\text{sg}}(x) = \begin{cases} 1 & \text{if } x = 0 \\ 0 & \text{if } x \neq 0 \end{cases}$

h. $x!$

i. $\min(x, y) = \text{minimum of } x \text{ and } y$

j. $\min(x_1, \dots, x_n)$

k. $\max(x, y) = \text{maximum of } x \text{ and } y$

l. $\max(x_1, \dots, x_n)$

m. $\text{rm}(x, y) = \text{remainder upon division of } y \text{ by } x$

n. $\text{qt}(x, y) = \text{quotient of division of } y \text{ by } x$

later example (just getting the tex down):

$$\tau(x) = \sum_{y \leq x} \overline{\text{sg}}(\text{rm}(y, x))$$

Examples

1. temp

2. temp 2
3. For $x > 0$, let ℓh be the number of nonzero exponents in the factorization of x into powers of primes. This is also the same as saying the number of distinct primes that divide x . Let $\ell h(0) = 0$. Then ℓh is primitive recursive.